

Job-shop scheduling with limited flexible workers considering ergonomic factors using an improved multi-objective discrete Jaya algorithm

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ABSTRACT

Job-Shop Scheduling Problem (JSP) with worker flexibility has gained significant attention in recent years, offering cost reduction and improved efficiency by limiting the number of workers. However, consideration of ergonomic factors is generally ignored for such systems. The ergonomic consideration is important to prevent stress, fatigue, and lack of focus among the workers, particularly when recurring manual tasks are involved. To prioritize productivity and workforce relief simultaneously, this article addresses the JSP with limited flexible workers considering ergonomic factors (JSPLFW-ER). The study focuses on worker workload and human energy expenditure as the key ergonomic factors. The average human energy expenditure rate is also considered to ensure that its value does not exceed the critical limit. An improved multi-objective discrete Jaya algorithm (IMDJA) is proposed to solve JSPLFW-ER by minimizing the makespan, maximum worker workload, and maximum human energy expenditure of the assigned operations. IMDJA incorporates a Jaya random partial updating mechanism in which a certain number of elements of the individual are randomly selected for Jaya updating. The number of elements to be selected for Jaya updating is evaluated using the Taguchi analysis. The performance of IMDJA is investigated with comprehensive experiments, using the 40 newly constructed instances and found competitive to solve JSPLFW-ER. Furthermore, a real-world industrial problem is presented with multiple experiments, which demonstrates the applicability of the JSPLFW-ER and effectiveness of the proposed IMDJA for solving the industrial problem.

1. Introduction

In the modern age of mass customization of products, the adaptability and adjustability of manufacturing resources due to rapid changes in product design, manufacturing process, production batch and lead time, is a great challenge. Production scheduling is an efficient way to undertake such demands, utilizing the available resources with a quick response. It has a significant influence on productivity and the cost of production. One of the most popular production scheduling in which each job has prescribed processing routes through machines is job-shop scheduling (JSS) (Zhang et al., 2017).

To improve the productivity of JSS and reduce the cost of workers, workers are trained to be flexible in handling different equipment and tools to undertake multiple tasks (Treleven, 1989). Worker flexibility can make the JSS more conducive, cost effective and avoid the management difficulty (Zhang et al., 2017), especially when the number of workers is less than the number of machines. Considering the machines and workers simultaneously in JSP is commonly called dual resource-

constrained JSP (DRCJSP). Typically, the DRCJSP consists of two tasks, i.e., sequencing of operations and allocating the workers to the corresponding machines from the set of capable workers.

Worker flexibility was introduced by Nelson (1967) for JSP. Fry et al. (1995) suggested that worker flexibility is an attractive choice to improve the manufacturing performance of the job-shop. It gives various strategic advantages and allows the company to buffer against any uncertain situation in production scheduling, and operation management activities. Yue et al. (2008) evaluated the JSS with worker flexibility and found that a flexible workforce enhances the system's responsiveness and reduces manufacturing lead time, decreases the level of work-in-process, and improves customer service. Li et al. (2016) and Zhong et al. (2018) presented a bi-objective solution for DRCJSP to optimize the makespan and processing cost. Thüerer et al. (2019) studied the performance of workload control in DRC high-variety shops with interchangeable workers by considering environmental aspects and various worker assignments and dispatching methods. Li et al. (2021) proposed a solution for many objectives of JSP with the limited workers

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utilizing the multi-skilled workforce to optimize makespan, total tardiness, total idle time, total worker cost, and total machine energy. Recently, [Dunke and Nickel \(2022\)](#) proposed a multi-method approach for scheduling and efficiency analysis in DRCJSP with processing time uncertainty. Worker flexibility is not limited to the JSP only; researchers have also employed worker flexibility in flexible job-shop scheduling problem (FJSP) ([Gnanavelbabu et al., 2021](#); [Gong et al., 2018a, 2021](#); [Lou et al., 2022](#); [Luo et al., 2023](#); [Luo et al., 2022](#)) and flow-shop scheduling problem (FSSP) ([Gong et al., 2020](#); [Hasani and Hosseini, 2022](#)). FJSP with worker flexibility is generally termed dual resource constrained flexible job-shop scheduling problem (DRCFJSP).

Besides worker flexibility, other human factors like physical fatigue, rest allowance, and learning-forgetting have also been researched by authors for DRCJSP. The learning and forgetting were investigated with physical fatigue recovery by [Givi et al. \(2015\)](#) for the DRCJSP. They found out that human performance affects system functioning and final product quality. Similarly, physical fatigue in DRCJSP considering workers' flexibility, fatigue, and recovery was evaluated by [Jaber and Neumann \(2010\)](#). The experimental results in this research revealed that the rest breaks after each task, short cycle times, and faster recovery rates improve the system's performance. The ageing workforce effects were also studied for DRCJSP considering fatigue through the rest allowance formulation in [Berti et al. \(2021\)](#). The existing research on DRCJSP mostly emphasized worker flexibility as a human factor with time and cost-associated objectives. Moreover, a few studies also considered other human factors but were only limited to learning-forgetting, fatigue recovery analysis, and rest allowance.

Due to the escalating labor costs, there is a growing interest in worker flexibility to enhance efficiency and decrease manufacturing costs by limiting the number of workers. In addition, in many real-world manufacturing systems, managers also have to operate with a limited number of workers (i.e. when the number of workers is less than the number of machines) due to occasional labor shortages without compromising productivity significantly. As a result, the need for worker flexibility is even more crucial in such situations. Contemporary manufacturers are increasingly inclined to hire and train workers with the ability to operate multiple types of equipment and tools in order to improve production efficiency and flexibility. To the best of our knowledge, only [Li et al. \(2021\)](#) considered the limited flexible workforce aspect in detail for DRCJSP. They presented multiple sub-instances for a combination of jobs and machines with a variable number of limited multiskilled workers. Nevertheless, their study did not consider the ergonomic factors such as maximum worker workload and human energy expenditure to relieve the workers. Balanced work assignment is important for the JSP with limited flexible workers to avoid the excessive work assigned to any worker to prevent stress, fatigue, lack of focus, and health issues, especially for the JSP in industries dealing with recurring manual tasks, such as marble workshop, casting workshop, and handicraft industry. Therefore, it is important to consider ergonomic factors to alleviate workers in such manufacturing conditions.

In this research, we propose a JSP with limited flexible workers considering ergonomic factors (JSPLFW-ER) to prioritize productivity and workers' relief simultaneously. The objective is to minimize the makespan, maximum worker workload, and maximum human energy expenditure of the assigned operations. Moreover, the average human energy expenditure rate is evaluated to ensure that its value does not exceed the critical limit. To the best of our knowledge, no existing studies addressed the JSP with limited flexible worker aspect considering the ergonomic factors, namely maximum worker workload and human energy expenditure.

JSP is an NP-hard problem, which has attracted much attention to the optimization methods to address the problem ([Gonçalves et al., 2005](#)). During recent years, *meta*-heuristics have been commonly applied to investigate the multi-objective JSP. [Zhao et al. \(2018\)](#) employed a two-generation Pareto ant colony algorithm for multi-objective JSP with alternative process plans and unrelated parallel

machines. [Zhao et al. \(2017\)](#) proposed an improved multi-objective evolutionary algorithm based on decomposition to optimize the makespan, tardiness time, and total flow time. [Huang and Yu \(2017\)](#) developed an effective ant colony optimization algorithm with equal-size lot-splitting to optimize the cost and makespan. [Abedi et al. \(2020\)](#) proposed a multi-objective memetic algorithm with a multi-population for energy-efficient JSP with deteriorating machines. In addition to the general local search, they also exploited an advanced object-oriented local search periodically on a particular portion of the population. [Wu et al. \(2020\)](#) introduced a predictive scheduling algorithm with buffering strategies to generate Pareto solutions for multiple objectives. The study optimizes risks in JSP under random machine breakdowns, considering makespan, makespan risk, and stability risk simultaneously. [Kouider and Ait Haddadène, \(2021\)](#) presented a bi-objective branch-&-bound and ϵ -constraint algorithms for the unit time JSP to minimize the makespan and stability risks. [Boukedroun et al. \(2023\)](#) proposed a hybrid algorithm combining a genetic algorithm and tabu search algorithm to solve the stochastic JSP with the objective to minimize the makespan and number of critical operations simultaneously.

In recent years, many multi-objective DRCJSP have also been solved using different *meta*-heuristics. [Li et al. \(2016\)](#) employed a branch population genetic algorithm for DRCJSP to minimize the makespan and cost. [Zhong et al. \(2018\)](#) presented a branch population genetic algorithm based on the compressed time window scheduling strategy employing the elite evaluation and a fan-shaped roulette operator for DRCJSP to minimize the makespan and cost. [Gong et al. \(2018a\)](#) produced a new memetic algorithm based on NSGAII using a three-layer chromosome demonstration and an efficient local search method for DRCFJSP to minimize the makespan, critical load, and total load of machines. [Gong et al. \(2018b\)](#) developed a new hybrid genetic algorithm for DRCFJSP and employed several efficient crossover and mutation operators for their problem to optimize the makespan, cost, and green factors. [Gong et al. \(2021\)](#) solved DRCFJSP through a non-dominated ensemble fitness ranking method using a critical path-based local search method to minimize the makespan, cost, and green factors. [Gnanavelbabu et al. \(2021\)](#) produced a backtracking search algorithm framework utilizing the integration of the Monte Carlo simulation for DRCFJSP to optimize the expected makespan and the standard deviation of makespan. [Wu et al. \(2021\)](#) solved the DRCFJSP through an improved NSGA-II considering fixtures' loading/unloading time to optimize the makespan and total setup time. [Peng et al. \(2022\)](#) developed a hybrid discrete multi-objective imperial competition algorithm to investigate the DRCFJSP in green sustainable manufacturing based on the learning effect. [Lou et al. \(2022\)](#) addressed the FJSP with consideration of human factors. They proposed a learning and decomposition-based multi-objective memetic algorithm, incorporating a learning-based adaptive local search into the multi-objective evolutionary algorithm based on decomposition (MOEA/D), to minimize the makespan, maximum machine workload, and total machine workload. [Luo et al. \(2022\)](#) investigated a distributed FJSP considering worker arrangement. They employed an improved memetic algorithm with an adaptive neighborhood search method to minimize the makespan, maximum workload of machines and maximum workload of workers. To minimize the same objectives for FJSP with worker cooperation flexibility, [Luo et al. \(2023\)](#) proposed a Pareto-based two-stage evolutionary algorithm. The algorithm includes a well-tailored initialization and NSGA-II for global exploration, and a competitive objective-based local search operator for improved local search ability.

The above literature reveals that many *meta*-heuristic algorithms have been exploited for multi-objective optimization while investigating the JSP and DRCJSP. These algorithms have some common limitations. They need algorithm specific parameters, need hybrid algorithms or local search methods to avoid the local optimum solution and obtain a better diversity of the solutions, making the implementation complex.

Jaya algorithm (JA) is a *meta*-heuristic proposed by [Venkata Rao \(2016\)](#) and it does not need algorithm specific parameters. It reduces the

parameter tuning efforts and can maintain a more stable performance. It is based on the concept that the solution moves towards the best solution and retracts from the worst solution. Due to these advantages, researchers' interest is growing in exploiting the JA to deal with single objective and multi-objective optimization problems in recent years. Accordingly, researchers have also applied the JA for job-shop scheduling problems. Caldeira and Gnanavelbabu (2019) solved the FJSP using an improved JA. Gao et al. (2019) proposed FJSP for new job insertion using the discrete JA. Li et al. (2020) produced an improved JA for solving the FJSP with transportation and setup times. Caldeira and Gnanavelbabu (2021) proposed a Pareto-based discrete JA to solve the multi-objective FJSP employing a neighborhood-based local search method and mutation operator. Fan et al. (2021) suggested a hybrid JA for solving FJSP considering multiple critical paths. He et al. (2021) developed an effective multi-objective JA to solve the JSP with an approach based on grey entropy parallel analysis and an opposition-based learning approach to obtain the enhanced results. Alawad and Abed-alguni (2022) proposed a discrete Jaya for the permutation flow-shop scheduling problem. They introduced five modifications in the discrete Jaya algorithm. The two significant modifications were employing refraction learning to generate a diverse initial population and using three different mutation methods. Xie et al. (2023) introduced a decomposition-based multi-objective Jaya algorithm to address lot-streaming JSP considering variable sublots and intermingling settings. They integrated problem-specific and neighborhood-based local search methods to improve the exploitation in the search process. Pan et al. (2023) addressed bi-objective distributed flow-shop scheduling problems with lot-streaming by utilizing an improved Jaya algorithm. They utilized a job-factory assignment strategy and introduced improved strategies for makespan and energy efficiency, enhancing the Jaya algorithm's performance.

Although the Jaya algorithm is applied in the JSP due to its advantages, however, it has the tendency like other *meta*-heuristics to get trapped at a local optimum solution after a few iterations and faces difficulty in attaining the optimal solution. Hybrid Jaya algorithms or local search methods are generally employed to overcome this issue (Caldeira and Gnanavelbabu, 2021; Fan et al., 2021; Li et al., 2020; Xie et al., 2023), which raises the implementation complexity.

In this research, an improved multi-objective discrete JA (IMDJA) is proposed to solve JSPLFW-ER. To avoid convergence to the local optimum solution and obtain a better diversity of the population, a Jaya random partial updating is proposed, where, instead of considering all the elements of the individual for Jaya updating as adopted in Caldeira and Gnanavelbabu (2021), a certain number of elements are randomly selected to apply Jaya updating. The number of elements of the individual to be selected for Jaya updating is evaluated using the Taguchi design of experiment (DOE) method (Montgomery, 2008).

The main contribution of this work is summarized as follows: JSP with limited flexible workers is addressed with consideration of ergonomic factors namely, the worker workload and the human energy expenditure. Minimization of the makespan, maximum worker workload, and maximum human energy expenditure have been simultaneously considered for JSP. An improved multi-objective discrete Jaya algorithm (IMDJA) is proposed to solve the JSPLFW-ER. The performance of IMDJA is compared with other algorithms employing 40 newly constructed instances and an industrial case study and found competitive.

The remainder of the paper is organized as follows: In Section 2, the problem description and mathematical model have been presented. Section 3 provides a brief description of the JA and the framework for the proposed IMDJA for solving JSPLFW-ER. The case study, experimental results, and discussions are presented in Section 4. Finally, the conclusion and future work are given in Section 5.

2. Problem description and mathematical modeling for JSPLFW-ER

2.1. Notations

The notations used in the JSPLFW-ER approach are given as follows:

n	Number of jobs
m	Number of machines
n_i	Number of operations of job i
l	Number of workers
i	Index of jobs
j	Index of operations
k	Index of machines
s	Index of workers
q	Index of postures
w	Index of micro-movements
O_{ij}	j^{th} operation of job i
u_{ij}	Number of body postures of a worker employed in operation O_{ij}
v_{ij}	Number of micro-movements of a worker in operation O_{ij}
M_{wij}	w^{th} micro-movement of a worker in operation O_{ij}
P_{qij}	q^{th} body posture of a worker employed in operation O_{ij}
STO_{ij}	Start time of operation O_{ij}
CTO_{ij}	Completion time of operation O_{ij}
CT_k	Completion time of machine k
T_{ijks}	Processing time of operation O_{ij} on machine k by worker s (min)
t_{wijks}	Time duration of micro-movement M_{wij} on machine k by worker s (min)
t_{qijks}	Time duration of body posture P_{qij} on machine k by worker s (min)
L_{wijks}	Load to move in micro-movement M_{wij} on machine k by worker s (Kg)
d_{wijks}	Horizontal displacement of the workpiece in micro-movement M_{wij} on machine k by worker s (meter)
F_{wijks}	Average applied force (pushing/pulling) by hands in micro-movement M_{wij} on machine k by worker s (Kg)
h_{1wijks}	Height from the floor; the starting point for lifting and endpoint for lowering of micro-movement M_{wij} on machine k by worker s (meter)
h_{2wijks}	Height from the floor; the starting point for lowering and endpoint for the lifting of micro-movement M_{wij} on machine k by worker s (meter)
GoS	Grade of the walking surface (%) (GoS is considered same during all operations)
G_{wijks}	Gender of worker s for micro-movement M_{wij} on machine k (for male $G = 1$, for female $G = 0$)
BW_{wijks}	Weight of worker s for micro-movement M_{wij} on machine k (Kg)
BW_{qijks}	Weight of worker s for posture P_{qij} on machine k (Kg)
V	Walking speed of worker (m/s) (V is considered same for all workers)
$\bar{\epsilon}_{qijks}$	Metabolic energy expenditure rate for maintaining the posture P_{qij} on machine k by worker s (Kcal/min)
$\bar{\epsilon}_{wijks}$	Energy expenditure of micro-movement M_{wij} on machine k by worker s (Kcal)
TO_s	Worker workload i.e. total time required to perform the assigned operations by worker s in a complete job-shop cycle (min)
TI_s	Total time of the idle state of worker s in a complete job-shop cycle (min)
EO_{ijks}	Human energy expenditure of operation O_{ij} on machine k by worker s (Kcal)
EO_s	Human energy expenditure of all the assigned operations by worker s in a complete job-shop cycle (Kcal)
EI_s	Human energy expenditure during idle time by worker s in a complete job-shop cycle (Kcal)
CT_M	Maximum completion time or makespan (min)
TO_M	Maximum TO_s (min)
EO_M	Maximum EO_s (min)
E_s	Total human energy expenditure by worker s in a complete job-shop cycle (Kcal)
$\bar{E}_s$	Average human energy expenditure rate by worker s in a complete job-shop cycle (Kcal/min)

Binary variables:

$$X_{ijk} = \begin{cases} 1, & \text{if operation } O_{ij} \text{ is done on machine } k \\ 0, & \text{else} \end{cases}$$

$$X_{ijks} = \begin{cases} 1, & \text{if operation } O_{ij} \text{ is done on machine } k \text{ by worker } s \\ 0, & \text{else} \end{cases}$$

$$X_{wijks} = \begin{cases} 1, & \text{if micro - movement } M_{wij} \text{ is done on machine } k \text{ by worker } s \\ 0, & \text{else} \end{cases}$$

$$X_{qijks} = \begin{cases} 1, & \text{if posture } P_{qij} \text{ is employed by worker } s \text{ on machine } k \\ 0, & \text{else} \end{cases}$$

Table 1

An instance of JSPLFW-ER ($4 \times 4 \times 3$) showing machine and set of capable workers against each machine (capable workers are ticked).

Job	Operation	Machine	Worker		
			W_1	W_2	W_3
J_1	O_{11}	M_1	✓	✓	✓
	O_{12}	M_2	–	✓	✓
	O_{13}	M_3	✓	–	✓
J_2	O_{21}	M_1	✓	✓	✓
	O_{22}	M_2	–	✓	✓
	O_{23}	M_4	✓	✓	–
J_3	O_{31}	M_3	✓	–	✓
	O_{32}	M_4	✓	✓	–
J_4	O_{41}	M_1	✓	✓	✓
	O_{42}	M_4	✓	✓	–

2.2. Problem description

The proposed JSPLFW-ER is an extension of JSP with limited flexible workers and consideration of ergonomic factors. The problem is described as follows: There are n jobs $J = \{J_1, J_2, \dots, J_n\}$ in the manufacturing system to be processed on m designated machines $M = \{M_1, M_2, \dots, M_m\}$ by a set of l flexible workers $W = \{W_1, W_2, \dots, W_l\}$. However, the number of workers is fewer than the number of machines for the JSPLFW-ER. A job J_i possesses a sequence of n_i operations $\{O_{i1}, O_{i2}, \dots, O_{in_i}\}$ to be performed consecutively as per the prescribed sequence. Each operation is performed on a specified machine. Moreover, an operation is performed by one worker selected from the set of capable workers. An example of a JSPLFW-ER is demonstrated in Table 1 with four jobs, four machines, and three workers ($4 \times 4 \times 3$). Each job has a certain number of operations with the designated machines and capable workers. The processing time of each operation is determined and depends on the worker processing it. The human energy expenditure of workers is calculated as per Garg's method (Garg et al. 1978) described in Section 2.3. Following are the assumptions for this problem:

- Each machine can only process one operation at one time.
- Worker can transfer from one machine to another after completing his/her task.
- No loss of productive time occurs when a worker is transferred from one machine to another.
- Each worker can operate only one machine at a time.
- An operation is carried out only once on one machine, and the operation sequence must be considered for each job.
- After starting an operation, a temporary interruption of an operation is not allowed.
- An operation must not be started until the completion of its preceding operations.
- There is no precedence constraint amongst the operation of different jobs.

2.3. Mathematical modeling

2.3.1. Human energy expenditure of an operation

Each operation consists of an operator's many micro-movements (walking, arm lowering, standing, etc.). The average human energy expenditure rate to perform an operation ($\overline{EO}$) combines two parts, i.e., the metabolic energy expenditure rate required to maintain the posture until completion of an operation and net metabolic energy expenditure for all the micro-movements to perform an operation. From Garg's method (Garg et al. 1978), we can derive the equation for the average human energy expenditure rate to perform a job operation O_{ij} on machine k by worker s (represented as $\overline{EO}_{ijks}$) as follows:

$$\overline{EO}_{ijks} = \frac{\sum_{q=1}^{u_{ij}} \bar{\xi}_{qijks} t_{qijks} X_{qijks} + \sum_{w=1}^{v_{ij}} \xi_{wijks} X_{wijks}}{T_{ijks}} \quad (1)$$

where $\bar{\xi}_{qijks}$ is the metabolic energy expenditure rate (Kcal/min) due to posture P_{qij} on machine k by worker s and it can be calculated as:

for standing:

$$\bar{\xi}_{qijks} = 0.024 BW_{qijks} \quad (2)$$

for sitting:

$$\bar{\xi}_{qijks} = 0.023 BW_{qijks} \quad (3)$$

for standing bent:

$$\bar{\xi}_{qijks} = 0.028 BW_{qijks} \quad (4)$$

$$T_{ijks} = \sum_{q=1}^{u_{ij}} t_{qijks} X_{qijks} \quad (5)$$

ξ_{wijks} is the net energy expenditure of micro-movement M_{wij} on machine k by worker s . It can be calculated through various Garg's model equations (Garg et al. 1978). Micro-movement equations for human energy expenditure employed in the case study of this article are provided in the following Eq. (6–15).

- For walking:

$$\xi_{wijks} = 10^{-2} [51 + 2.54 BW_{wijks} \cdot V^2 + 0.379 BW_{wijks} \cdot GoS \cdot V] t_{wijks} \quad (6)$$

- For squat lift:

$$\xi_{wijks} = 10^{-2} [0.514 BW_{wijks} (0.81 - h_{1wijks}) + (2.19 L_{wijks} + 0.62 G_{wijks} \cdot L_{wijks}) (h_{2wijks} - h_{1wijks})] \quad (7)$$

- For carrying loads held against the thighs or the waist

$$\xi_{wijks} = 10^{-2} [68 + 2.54 BW_{wijks} \cdot V + 4.08 L_{wijks} \cdot V^2 + 11.4 L_{wijks} + 0.379 (L_{wijks} + BW_{wijks}) GoS \cdot V] t_{wijks} \quad (8)$$

- For arm lifting:

$$\xi_{wijks} = 10^{-2} [0.062 \cdot BW_{wijks} (h_{2wijks} - 0.81) + (3.19 L_{wijks} - 0.52 G_{wijks} \cdot L_{wijks}) (h_{2wijks} - h_{1wijks})] \quad (9)$$

- For holding at arm's length (both hands)

$$\xi_{wijks} = 0.037 L_{wijks} t_{wijks} \quad (10)$$

- For pushing/pulling, at bench height (1.5 m) (Kcal/push)

$$\xi_{wijks} = 10^{-2} d_{wijks} (0.112 BW_{wijks} + 1.15 F_{wijks} + 0.505 G_{wijks} F_{wijks}) \quad (11)$$

- For arm lowering:

$$\xi_{wijks} = 10^{-2} [0.093 BW_{wijks} (h_{2wijks} - 0.81) + (1.02 L_{wijks} + 0.37 G_{wijks} \cdot L_{wijks}) (h_{2wijks} - h_{1wijks})] \quad (12)$$

- For squat lowering:

$$\xi_{wijks} = 10^{-2} [0.511 BW_{wijks} (0.81 - h_{1wijks}) + 0.701 L_{wijks} (h_{2wijks} - h_{1wijks})] \quad (13)$$

- For lateral movement of arms of 90 degrees, sitting, one hand (Kcal/movement)

$$\xi_{wijks} = 10^{-2}(2.54 + 1.1 L_{wijks} + 0.248 G_{wijks} \cdot L_{wijks}) \quad (14)$$

- For forward movement of arms, sitting, one or both hands (Kcal/movement)

$$\xi_{wijks} = 10^{-2} d_{wijks} (6.3 + 2.71 L_{wijks}) \quad (15)$$

From Eq. (1), the total human energy expenditure to perform an operation O_{ij} on machine k by worker s is given by:

$$EO_{ijks} = \sum_{q=1}^{u_{ij}} \bar{\xi}_{qijks} t_{qijks} X_{qijks} + \sum_{w=1}^{v_{ij}} \xi_{wijks} X_{wijks} \quad (16)$$

2.3.2. Human energy expenditure for a job-shop cycle

The total human energy expenditure by worker s for a complete job-shop cycle (E_s) is the combination of two parts:

1. Energy expenditure of operations by worker s (EO_s) to perform the assigned operations during a job-shop cycle. It is calculated as:

$$EO_s = \sum_{i=1}^n \sum_{j=1}^{n_i} \sum_{k=1}^m (EO_{ijks} X_{ijks}) \quad (17)$$

2. Energy expenditure of idle state by worker s (EI_s), i.e., when a worker is not performing any job operation and the energy expenditure is only due to maintaining the posture. It is calculated as:

$$EI_s = \bar{\xi}_s \cdot TI_s \quad (18)$$

where $\bar{\xi}_s$ is the metabolic energy expenditure rate (Kcal/min) due to maintenance of posture of worker s . We consider that the worker is standing in the idle state, hence, $\bar{\xi}_s = 0.024 BW_s$, where BW_s is the body weight of worker s . TI_s is the total idle time of worker s during the job-shop cycle when s /he does not perform any operation and waiting for the job. It is calculated as:

$$TI_s = CT_M - TO_s \quad (19)$$

where CT_M is the maximum completion time or makespan and TO_s is the worker workload i.e. total time required to perform the assigned operations by worker s in a job-shop cycle, TO_s is calculated as follows:

$$TO_s = \sum_{i=1}^n \sum_{j=1}^{n_i} \sum_{k=1}^m (T_{ijks} X_{ijks}) \quad (20)$$

2.3.3. Average human energy expenditure rate for a job-shop cycle

The average human energy expenditure rate of a worker s for a complete job-shop can be calculated using the following formula:

$$\bar{E}_s = \frac{EO_s + EI_s}{CT_M} \quad (21)$$

where $\bar{E}_s$ is the average human energy expenditure rate of a worker s for a complete job-shop cycle.

The critical limit of the average human energy expenditure rate for an eight-hour working period for a young and healthy male is recommended as 5.2 Kcal/min (Garg et al. 1978). However, in real-world manufacturing, variably aged and mixed-gender workers perform the tasks. Therefore, the critical limit of the average human energy expenditure rate ($\bar{E}_{lim}$) can be set according to the workers' conditions and working environment.

2.3.4. Modeling for objective functions

Based on the above equations and assumptions, the mathematical model of the objective functions for JSPLFW-ER is given as:

1. Minimization of makespan CT_M

$$\text{Min } CT_M = \max_{1 \leq k \leq m} (CT_k) \quad (22)$$

2. Minimization of maximum worker workload TO_M

$$\text{Min } TO_M = \max_{1 \leq s \leq l} (TO_s) \quad (23)$$

3. Minimization of maximum human energy expenditure EO_M

$$\text{Min } EO_M = \max_{1 \leq s \leq l} (EO_s) \quad (24)$$

The constraints and descriptions are given as follows:

- The operation precedence constraint is given by the following equation:

$$CTO_{ij} - CTO_{i(j-1)} \geq \sum_{k=1}^m \sum_{s=1}^l T_{ijks} X_{ijks} \quad (25)$$

where $\forall i, j : i = 1, \dots, n; j = 2, \dots, n_i$

- The relationship between starting and completion time of an operation is:

$$CTO_{ij} = STO_{ij} + \sum_{k=1}^m \sum_{s=1}^l T_{ijks} X_{ijks} \quad (26)$$

where $\forall i, j : i = 1, \dots, n; j = 1, \dots, n_i$

- The machine limitation is that at one time each machine can process only one operation:

$$\sum_{i=1}^n \sum_{j=1}^{n_i} X_{ijk} = 1 \quad (27)$$

where $\forall k : k = 1, \dots, m$

- The worker restriction that at one time each worker can process only one operation:

$$\sum_{i=1}^n \sum_{j=1}^{n_i} X_{ijks} = 1 \quad (28)$$

where $\forall k, s : k = 1, \dots, m; s = 1, \dots, l$

- The process constraint that each operation is processed only once:

$$\sum_{k=1}^m \sum_{s=1}^l X_{ijks} = 1 \quad (29)$$

where $\forall i, j : i = 1, \dots, n; j = 1, \dots, n_i$

- The restriction that for an operation only one machine is chosen:

$$\sum_{k=1}^m X_{ijk} = 1 \quad (30)$$

where $\forall i, j : i = 1, \dots, n; j = 1, \dots, n_i$

O_{11}	O_{12}	O_{21}	O_{31}	O_{32}	O_{13}	O_{22}	O_{23}	O_{41}	O_{42}	
1	1	2	3	3	1	2	2	4	4	OSV
M_1	M_2	M_1	M_3	M_4	M_3	M_2	M_4	M_1	M_4	
1	2	1	3	4	3	2	4	1	4	MAV
W_1	W_2	W_2	W_3	W_2	W_1	W_3	W_2	W_3	W_1	
1	2	2	3	2	1	3	2	3	1	WAV

Fig. 1. Encoding example for the solution of JSPLFW-ER.

- The constraint that only one worker is selected to operate one machine:

$$\sum_{s=1}^I X_{ijk_s} = 1 \quad (31)$$

where $\forall i, j, k : i = 1, \dots, n; j = 1, \dots, n_i; k = 1, \dots, m$

- The constraint that the average energy expenditure rate of worker s for a complete job-shop cannot exceed the critical limit $\bar{E}_{lim}$ is ensured by the following equation:

$$\bar{E}_s = \frac{EO_s + EI_s}{CT_M} \leq \bar{E}_{lim} \quad (32)$$

3. Proposed IMDJA algorithm for JSPLFW-ER

3.1. A brief introduction to classical Jaya algorithm

The Jaya algorithm (JA) was developed by Venkata Rao (2016). The methodology of the JA describes that the solution to the given problem has to move toward the best available solution and goes away from the worst known solution. A brief description of the JA is given below:

- Set initial parameters such as population size, and the termination criterion.
- Generate the initial population and find out the solutions to objective functions.
- Identify the worst and the best solution for the population.
- Modify the solutions based on the best and the worst solution with the following equation:

$$X_{new,i} = X_i + r_1(X_{best} - |X_i|) - r_2(X_{worst} - |X_i|), \quad i = 1, 2, \dots, N \quad (33)$$

where $i = 1, 2, \dots, N$ (N is the population size); $X_{new,i}$ is the updated value of the corresponding initial individual X_i ; and r_1 and r_2 are two uniform random numbers in the range $[0,1]$.

- Compare the updated individual $X_{new,i}$ and initial individual X_i . If the objective value of the updated individual is better than the initial individual, replace it with the updated individual; otherwise, retain the initial solution. The resulting population is then considered for the next iteration.

3.2. Coding mechanism for JSPLFW-ER

The solution of the JSPLFW-ER is represented through a three-layer vector encoding, the operation process sequence vector (OSV), the machine assignment vector (MAV), and the worker allocation vector (WAV). Using the data given in Table 1, the three-layer vector encoding is demonstrated in Fig. 1. The OSV consists of all the operations performed in a complete production cycle integrating the precedence constraints, i.e. the operation O_{ij} can only be started after completing its immediate job predecessor $O_{i(j-1)}$. The numbers from left to right indicate the job number, while successive occurrences indicate the corresponding operation sequence of the job. Moreover, MAV shows the designated machines corresponding to each operation. The WAV represents the capable workers against the respective machines. The length of MAV and WAV is equal to OSV. The initial population for OSV is generated randomly. The machine designated for each corresponding operation in OSV is placed in a new vector to form MAV. Moreover, a worker to operate a machine is randomly selected from the capable worker set of the corresponding machine; consequently, WAV is generated.

Based on the above encoding, decoding can be obtained by one OSV, one MAV and one WAV with the active decoding method proposed by Gong et al. (2018a).

3.3. Jaya random partial updating mechanism in proposed IMDJA

JA was introduced to solve continuous optimization problems. However, a discrete JA (DJA) is considered appropriate to solve combinatorial optimization problems (Caldeira and Gnanavelbabu, 2021). This research proposes an improved multi-objective discrete Jaya

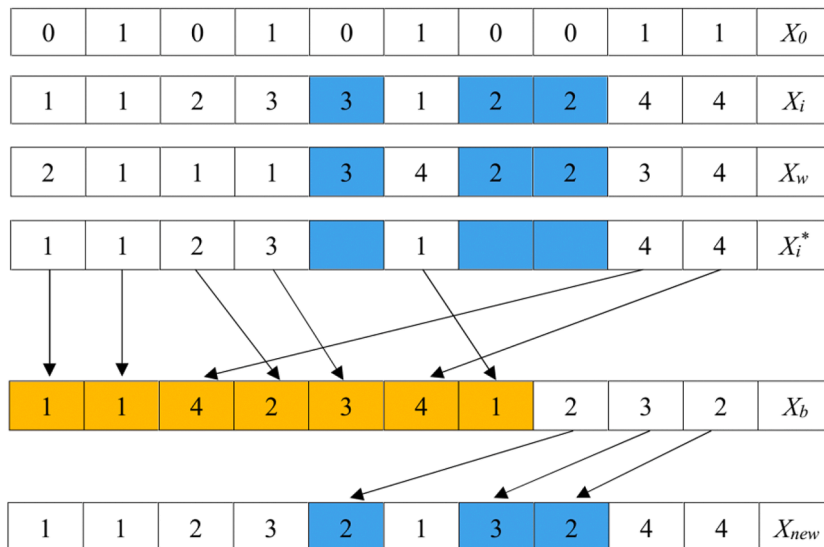


Fig. 2. JRPU mechanism for operation sequence.

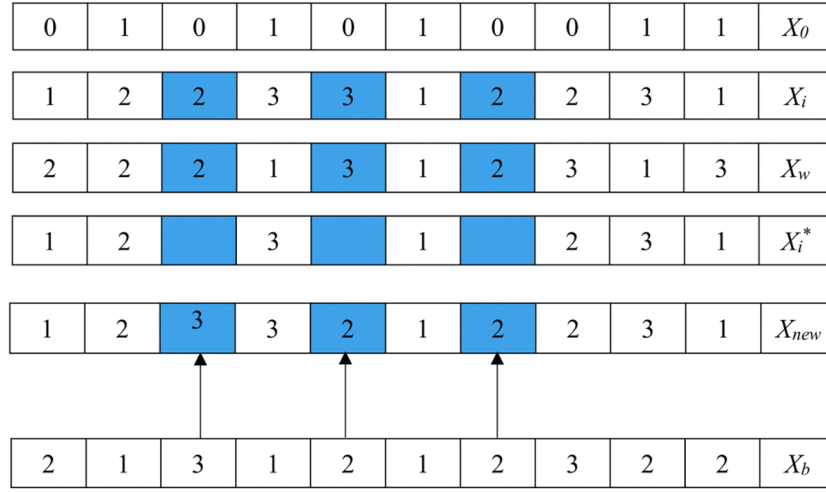


Fig. 3. JRPU mechanism for worker allocation.

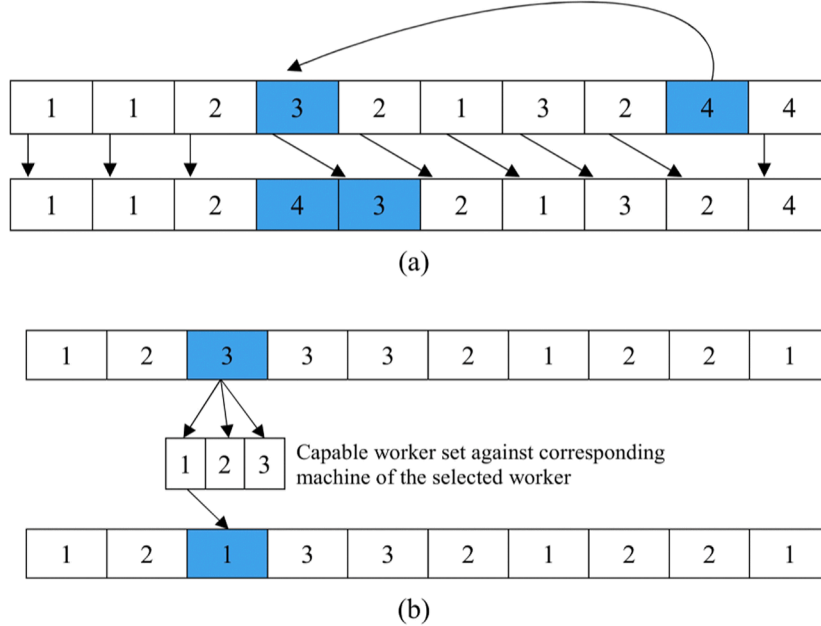


Fig. 4. Mutation operation for (a) OS and (b) WA.

algorithm (IMDJA) to solve the JSPLFW-ER. Instead of considering all the elements of the initial individual for Jaya updating as adopted in [Caldeira and Gnanavelbabu \(2021\)](#), a Jaya random partial updating (JRPU) mechanism is proposed in this article. Consequently, a certain number of elements of the initial individual are randomly selected to apply the Jaya updating. Moreover, the non-dominated sorting method ([Deb et al., 2002](#)) is employed to push the solutions toward the Pareto front (PF). Detail of the JRPU is given in the following paragraphs.

3.3.1. Operation sequence updating

The procedure description of the JRPU for operation sequence is given below:

- (1) Randomly select an initial individual X_i from the set of N population.
- (2) Determine the objective function values of all the population and identify the best and worst values based on non-dominated

sorting (discussed in [Section 3.6](#)). The corresponding best and worst individuals, i.e., X_b and X_w are extracted.

- (3) Generate a binary vector X_0 of the same length as X_i , consisting of randomly placed '0' and '1' elements. The number of zeros in X_0 is called the number of Jaya elements ($NoJE$), i.e., Jaya updating is applied only to the elements of X_i corresponding to zeros of X_0 . The value of $NoJE$ is decided through parameter settings (discussed in [Section 4.4](#)).
- (4) The new vector X_i^* is generated such that it receives the elements of the initial individual X_i at positions corresponding to 1 of X_0 . For the position of X_i^* corresponding to the 0 of X_0 , if there are same elements in X_i corresponding to X_w then these positions are left vacant; otherwise filled with the corresponding elements of X_i . In this way, a new vector X_i^* is created.
- (5) The elements of X_i^* are compared with X_b one by one from left to right, and the same elements shaded in X_b are removed. The remaining (unshaded) elements of X_b are placed in the vacant

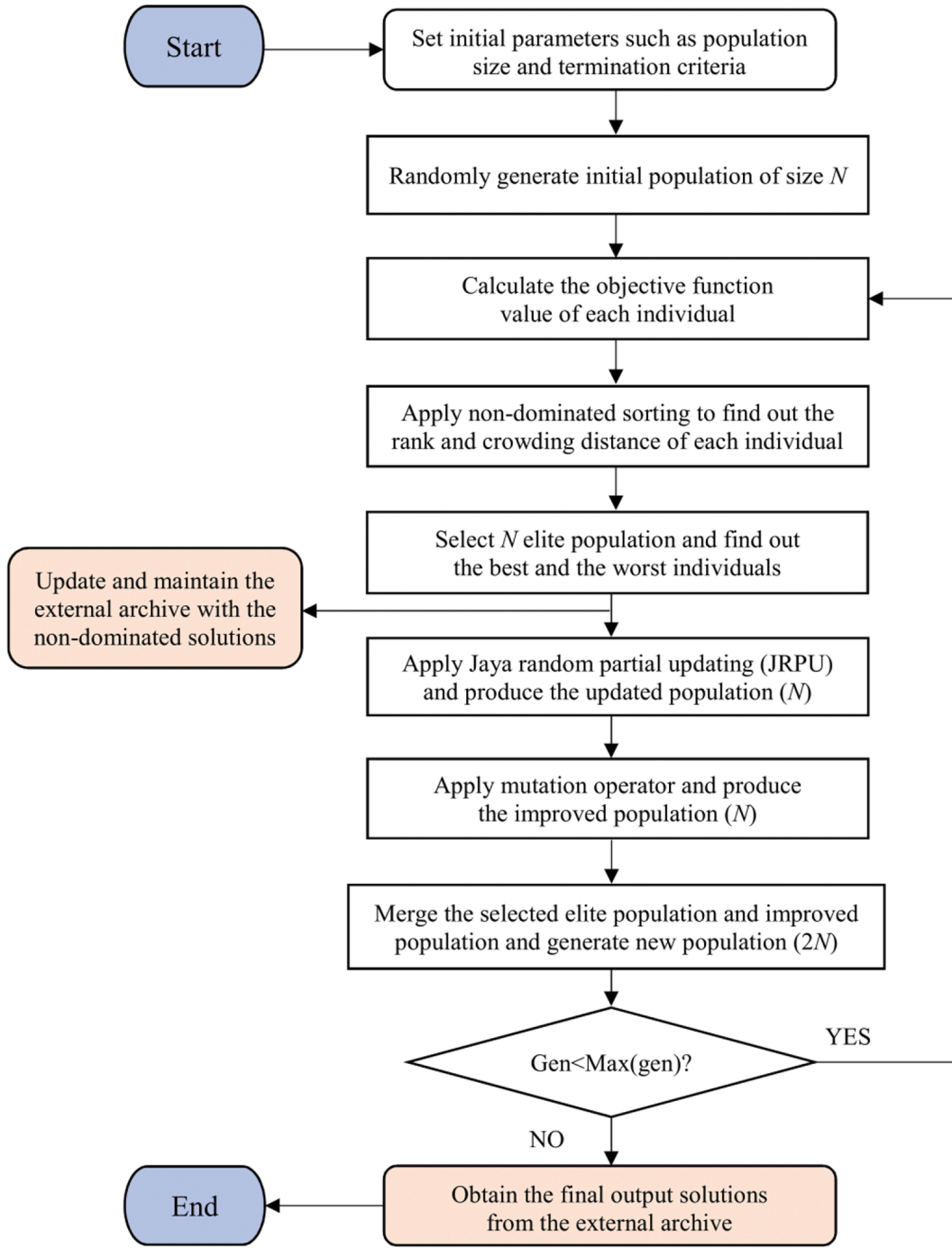


Fig. 5. Flowchart for IMDJA to solve JSPLFW-ER.

positions of X_i^* in the corresponding order. In this way, a new individual X_{new} is generated through JRPJ.

The illustration of JRPJ for operation sequence (with $NoJE = 5$) is shown in Fig. 2.

3.3.2. Worker allocation updating

The procedure description of the JRPJ for worker allocation updating is given below:

- (1) Select the individual X_i with the updated operation sequence.
- (2) Extract the best and worst individuals, i.e. X_b and X_w .

- (3) Using the binary vector X_0 , generate a new vector X_i^* such that it receives the elements of the initial individual X_i at positions corresponding to elements of 1 of X_0 . For the elements corresponding to 0 of X_0 , if there are same elements in X_i corresponding to X_w , these positions are left vacant; otherwise filled with the corresponding elements of X_i . Accordingly, a vector X_i^* is generated.

- (4) Each vacant element of X_i^* is filled with the corresponding element of X_b . In this way, a new individual X_{new} is generated.

The process of the worker allocation updating mechanism is shown in Fig. 3.

Table 2
Factor level for key parameters.

Parameter	Factor Level			
	1	2	3	4
N	50	100	150	200
$RoJE$	0.01	0.05	0.1	0.15

3.4. Mutation

The mutation is performed to further enhance the diversity of the population. For OS and WA, two mutation operators are employed as described below:

Mutation of OS: Two elements of the OSV are randomly selected. The first element is inserted before the second element.

Mutation of WA: An individual is randomly selected in WAV and replaced with the randomly selected worker from the set of capable workers of the corresponding machine.

The mutation operation for OS and WA is shown in Fig. 4 and the overall mutation procedure is described as follows:

- (1) Mutation probability p_m is defined to select a certain part of the total population.
- (2) An individual is randomly selected from the population.
- (3) A random number r_m , ranging from 0 to 1, is generated.
- (4) If $r_m < p_m$, the mutation process continues in the next step; otherwise, proceeds to the end.
- (5) Two different elements OE_1 and OE_2 are randomly selected in OSV and a random element WE_1 is selected in the corresponding WAV.
- (6) For OSV, the element OE_1 is inserted before OE_2 as shown in Fig. 4 (a).
- (7) For WAV, WE_1 is replaced with another worker randomly selected from the set of capable workers of the corresponding machine as shown in Fig. 4(b).
- (8) End.

3.5. Selection of population based on non-dominated sorting

Non-dominated sorting-based selection (Deb et al., 2002) of the population is performed as per the following steps:

- (1) Improved population (N) obtained after Jaya random partial updating and mutation is merged with the initial population (N) so that the total new population become $2N$.

- (2) After calculating the objective functions of each individual, non-dominated sorting is implemented, and the individuals of the population ($2N$) are ranked from low to high into different sets $R_1, R_2, R_3, \dots, R_n$.
- (3) Solutions obtained in a smaller rank are better than those in a larger rank, but those obtained in the same rank are distinguished by an additional criterion called crowding distance. The solutions having greater crowding distance values are considered better.
- (4) Finally, N elite individuals are selected based on the ranks and crowding distance, for the next generation.

3.6. Selection of the best and worst solutions

The best and worst individuals are selected among the population obtained through non-dominated sorting. The best individual is chosen randomly from the rank-1, whereas the worst individual is chosen randomly from the worst rank.

3.7. External archive generation

External archive generation is frequently employed in multi-objective algorithms to store elitist solutions. For IMDJA, the external archive is generated using the non-dominated sorting method (Deb et al., 2002). After each iteration, the non-dominated solutions (rank-1 solutions) are stored in the external archive. In order to enhance the quality of the solutions, non-dominated sorting is applied after each iteration to maintain only the non-dominated solutions in the external archive. When the termination condition is met, the solutions in the external archive will be the final output.

3.8. Flowchart of the proposed IMDJA to solve JSPLFW-ER

The flowchart of methodologies discussed above for the proposed IMDJA to solve the JSPLFW-ER is presented in Fig. 5.

4. Experimental results and analysis

4.1. Generation of test instances

In order to validate the performance of the proposed IMDJA for JSPLFW-ER, 40 instances are generated for various sizes of the problems with respect to the number of jobs (n), and number of machines (m). The combination $n \times m$ is used to represent the size of the problem. The combinations employed in this problem are 6×6 , 10×5 , 10×10 , 15×5 , 15×10 , 20×5 , 20×10 , 30×5 , 30×10 , 50×10 and 100×10 . The basic processing time bT_{ijk} of operation O_{ij} on machine k is obtained

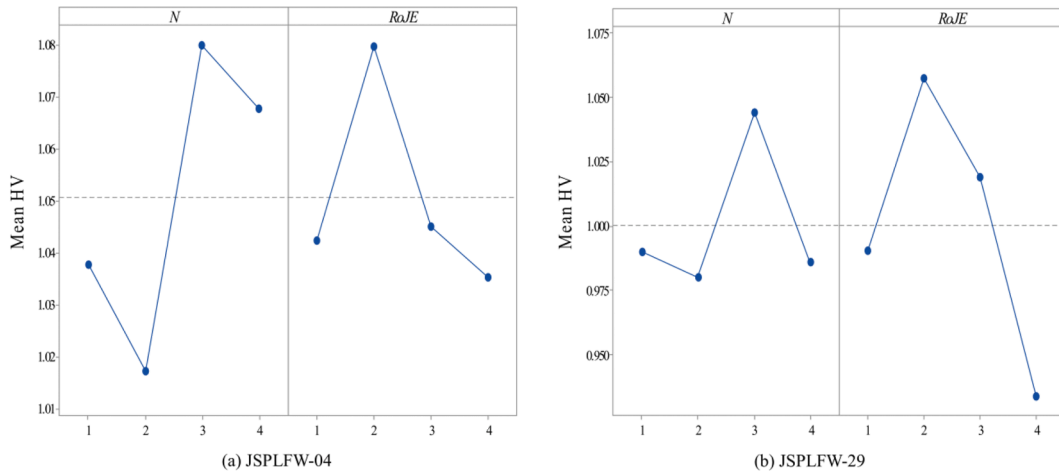


Fig. 6. Factor level trend of key parameters for the selected instances.

Table 3

Mean value of IGD and HV obtained by IMDJA under JRPU, JAEU and JUU.

Instance	$n \times m \times l$	IGD			-	HV		
		JRPU	JAEU	JUU		JRPU	JAEU	JUU
JSPLFW-01	$6 \times 6 \times 2$	0.0000	0.4162	0.4632		1.2934	0.5960	0.6120
JSPLFW-02	$6 \times 6 \times 3$	0.0000	0.3532	0.3734		1.1982	0.5361	0.5880
JSPLFW-03	$10 \times 5 \times 2$	0.0000	0.3487	0.4333		1.2672	0.5721	0.4984
JSPLFW-04	$10 \times 5 \times 3$	0.0000	0.4235	0.3575		1.1810	0.4654	0.5470
JSPLFW-05	$10 \times 10 \times 2$	0.0000	0.5951	0.5007		1.2620	0.4107	0.4653
JSPLFW-06	$10 \times 10 \times 3$	0.0000	0.4972	0.3255		1.1559	0.3284	0.5615
JSPLFW-07	$10 \times 10 \times 4$	0.0000	0.5386	0.4231		1.2247	0.3439	0.4495
JSPLFW-08	$10 \times 10 \times 5$	0.0000	0.6958	0.6080		1.2050	0.1735	0.2665
JSPLFW-09	$10 \times 10 \times 6$	0.0000	0.3474	0.2620		0.9058	0.3172	0.4332
JSPLFW-10	$15 \times 5 \times 2$	0.0000	0.7885	0.7867		1.2599	0.1844	0.1907
JSPLFW-11	$15 \times 5 \times 3$	0.0000	1.2522	0.3276		1.3041	0.0117	0.6704
JSPLFW-12	$15 \times 10 \times 2$	0.0000	0.3673	0.3044		1.1560	0.4515	0.5149
JSPLFW-13	$15 \times 10 \times 3$	0.0000	0.4079	0.3304		1.1980	0.4490	0.5654
JSPLFW-14	$15 \times 10 \times 4$	0.0000	0.3639	0.4451		1.1709	0.4754	0.4352
JSPLFW-15	$15 \times 10 \times 5$	0.0000	0.3848	0.3117		1.2074	0.4596	0.6214
JSPLFW-16	$15 \times 10 \times 6$	0.0000	0.3396	0.2410		0.9953	0.4384	0.5581
JSPLFW-17	$20 \times 5 \times 2$	0.0000	0.6089	0.5203		1.1758	0.2322	0.3405
JSPLFW-18	$20 \times 5 \times 3$	0.0000	0.6257	0.5247		1.3012	0.3540	0.4263
JSPLFW-19	$20 \times 10 \times 2$	0.0155	0.2892	0.3311		0.9907	0.6049	0.4198
JSPLFW-20	$20 \times 10 \times 3$	0.0000	0.4067	0.3301		1.1701	0.4161	0.5031
JSPLFW-21	$20 \times 10 \times 4$	0.0000	0.3688	0.2938		1.1335	0.4680	0.5827
JSPLFW-22	$20 \times 10 \times 5$	0.0000	0.4105	0.3197		1.1624	0.3967	0.5424
JSPLFW-23	$20 \times 10 \times 6$	0.0000	0.3732	0.3004		1.0444	0.3571	0.4609
JSPLFW-24	$30 \times 5 \times 2$	0.0000	0.3268	0.3045		1.1439	0.4822	0.5414
JSPLFW-25	$30 \times 5 \times 3$	0.0000	0.3821	0.3512		1.1604	0.4395	0.5102
JSPLFW-26	$30 \times 10 \times 2$	0.0000	0.5777	0.5432		1.2342	0.3345	0.3442
JSPLFW-27	$30 \times 10 \times 3$	0.0000	0.6201	0.4466		1.1397	0.1820	0.2944
JSPLFW-28	$30 \times 10 \times 4$	0.0000	0.5939	0.4701		1.1554	0.2241	0.3110
JSPLFW-29	$30 \times 10 \times 5$	0.0000	0.4509	0.2981		1.2423	0.4344	0.6302
JSPLFW-30	$30 \times 10 \times 6$	0.0000	0.4703	0.3217		0.9158	0.2312	0.3728
JSPLFW-31	$50 \times 10 \times 2$	0.0000	0.6840	0.6414		1.1913	0.1772	0.2225
JSPLFW-32	$50 \times 10 \times 3$	0.0000	0.5263	0.4600		1.1550	0.2517	0.3279
JSPLFW-33	$50 \times 10 \times 4$	0.0000	0.3806	0.3525		1.2172	0.4953	0.5670
JSPLFW-34	$50 \times 10 \times 5$	0.0000	0.4423	0.3680		1.1779	0.3619	0.4614
JSPLFW-35	$50 \times 10 \times 6$	0.0000	0.2887	0.2821		0.9959	0.4620	0.4928
JSPLFW-36	$100 \times 10 \times 2$	0.0000	0.5880	0.6336		1.1400	0.2135	0.1928
JSPLFW-37	$100 \times 10 \times 3$	0.0000	0.6137	0.5897		1.1908	0.2352	0.2530
JSPLFW-38	$100 \times 10 \times 4$	0.0000	0.5149	0.4224		1.2313	0.3293	0.4580
JSPLFW-39	$100 \times 10 \times 5$	0.0000	0.4659	0.4151		1.1410	0.3417	0.3979
JSPLFW-40	$100 \times 10 \times 6$	0.0000	0.2757	0.2653		1.0517	0.5179	0.5386

using the uniform distribution U [30 120] for each combination. Each combination is further divided based on the number of workers (l) ranges between $2 \leq l \leq [0.6m]$ (where, $l = 2, 3, 4, \dots$). In this problem, the total flexibility of workers is considered for all the instances i.e., each worker can manipulate all the machines. Accordingly, instances JSPLFW-01 to JSPLFW-40 are generated e.g., for combination 6×6 , the instances $6 \times 6 \times 2$ and $6 \times 6 \times 3$ are generated, where 2 and 3 indicate the number of workers. The actual processing time T_{ijks} of operation O_{ij} on machine k by workers s is generated using the normal distribution function in which the mean value is equal to the basic processing time (bT_{ijk}) with standard deviation of $0.2 \cdot bT_{ijk}$. The value of human energy expenditure is calculated against each process by each worker. The processing time is multiplied by a randomly selected average human energy expenditure rate from the set of $EO = \{3.8 \ 4.8 \ 5.8\}$. The data of the instances is provided as 'Appendix-A' on https://www.researchgate.net/publication/374417627_Appendices-ABC with extraction code: asir 49.

4.2. Performance metrics

The performance of the IMDJA algorithm for solving JSPLFW-ER is evaluated through three different performance metrics, namely inverted generational distance (IGD) (Yuan and Xu, 2015), hypervolume (HV) (Zitzler and Thiele, 1999) and Set Coverage (Yuan and Xu, 2015).

4.2.1. Inverted generational distance (IGD)

The IGD is widely exploited to evaluate the distribution, convergence, and diversity of a solution set. It is calculated as:

$$IGD(P, T^*) = \frac{1}{|T^*|} \sum_{u \in T^*} \min_{v \in P} d(u, v) \quad (34)$$

where P is the Pareto front (PF) obtained by an algorithm, T^* is the true PF, $d(u, v)$ indicates the Euclidean distance calculated between the points u and v , and $|T^*|$ shows the size of T^* . Moreover, T^* is the set of non-dominated solutions among the final solutions obtained by all algorithms in all runs. The smaller value of $IGD(P, T^*)$ means P is closer to T^* and shows better performance. Before evaluating the IGD, the values of non-dominated solutions of all the algorithms and the T^* are normalized as:

$$\bar{f}_i(x) = \frac{f_i(x) - f_i^{\min}}{f_i^{\max} - f_i^{\min}}, \quad i = 1, 2, 3 \quad (35)$$

where $f_i^{\max}$ and $f_i^{\min}$ are the maximal and minimal values of the i^{th} objective function, respectively, among all solutions.

4.2.2. Hypervolume (HV)

The HV measures the volume enclosed by the solutions in the PF obtained by the algorithm and a reference point. It is measured as:

$$HV(X, r) = \text{volume} \left(\bigcup_{x \in X} [f_1(x), r_1] \times \dots \times [f_\beta(x), r_\beta] \right) \quad (36)$$

Table 4

Mean value of SC obtained by IMDJA under JRPJ, JAEU and JUJ.

Instance	SC (JRPJ, JAEU)	SC (JAEU, JRPJ)	SC (JRPJ, JUJ)	SC (JUJ, JRPJ)
JSPLFW-01	1.0000	0.0000	1.0000	0.0000
JSPLFW-02	1.0000	0.0000	1.0000	0.0000
JSPLFW-03	1.0000	0.2568	1.0000	0.0000
JSPLFW-04	1.0000	0.0000	1.0000	0.0000
JSPLFW-05	1.0000	0.0000	1.0000	0.0000
JSPLFW-06	1.0000	0.0000	1.0000	0.0000
JSPLFW-07	1.0000	0.0000	1.0000	0.0000
JSPLFW-08	1.0000	0.0000	1.0000	0.0000
JSPLFW-09	1.0000	0.0000	1.0000	0.0000
JSPLFW-10	1.0000	0.0000	1.0000	0.0000
JSPLFW-11	1.0000	0.0000	1.0000	0.0000
JSPLFW-12	1.0000	0.0000	1.0000	0.0000
JSPLFW-13	1.0000	0.0000	1.0000	0.0000
JSPLFW-14	1.0000	0.0000	1.0000	0.0000
JSPLFW-15	1.0000	0.0000	1.0000	0.0000
JSPLFW-16	1.0000	0.0000	1.0000	0.0000
JSPLFW-17	1.0000	0.0000	1.0000	0.0000
JSPLFW-18	1.0000	0.0000	1.0000	0.0000
JSPLFW-19	0.4285	0.0112	1.0000	0.0000
JSPLFW-20	1.0000	0.0000	1.0000	0.0000
JSPLFW-21	1.0000	0.0000	1.0000	0.0000
JSPLFW-22	1.0000	0.0000	1.0000	0.0000
JSPLFW-23	1.0000	0.0000	1.0000	0.0000
JSPLFW-24	1.0000	0.0000	1.0000	0.0000
JSPLFW-25	1.0000	0.0000	1.0000	0.0000
JSPLFW-26	1.0000	0.0000	1.0000	0.0000
JSPLFW-27	1.0000	0.0000	1.0000	0.0000
JSPLFW-28	1.0000	0.0000	1.0000	0.0000
JSPLFW-29	1.0000	0.0000	1.0000	0.0000
JSPLFW-30	1.0000	0.0000	1.0000	0.0000
JSPLFW-31	1.0000	0.0000	1.0000	0.0000
JSPLFW-32	1.0000	0.0000	1.0000	0.0000
JSPLFW-33	1.0000	0.0000	1.0000	0.0000
JSPLFW-34	1.0000	0.0000	1.0000	0.0000
JSPLFW-35	1.0000	0.0000	1.0000	0.0000
JSPLFW-36	1.0000	0.0000	1.0000	0.0000

Table 4 (continued)

Instance	SC (JRPJ, JAEU)	SC (JAEU, JRPJ)	SC (JRPJ, JUJ)	SC (JUJ, JRPJ)
JSPLFW-37	1.0000	0.0000	1.0000	0.0000
JSPLFW-38	1.0000	0.0000	1.0000	0.0000
JSPLFW-39	1.0000	0.0000	1.0000	0.0000
JSPLFW-40	1.0000	0.0000	1.0000	0.0000

where $r = r_1, r_2, \dots, r_p$ indicates a reference point in the objective space dominated by all the solutions in X , and the HV shows the volume generated by r and the solutions in X . We use the normalized values of the solutions of all algorithms and the r is set to (1.1, 1.1, 1.1). The larger value of HV shows better convergence and diversity of the solutions.

4.2.3. Set Coverage (SC)

The quality of solutions is commonly measured through the SC. It is described as:

$$SC(X, Y) = \frac{|\{a \in Y \mid \exists b \in X : b \text{ dominates } a\}|}{|Y|} \quad (37)$$

where X and Y indicate the two solution sets, $|Y|$ shows the total solutions in Y . $SC(X, Y)$ measures the fractions of members of Y that are dominated by members of X . $SC(X, Y) = 1$ indicates that all solutions in Y are dominated by at least one solution in X . Whereas, $SC(X, Y) = 0$ demonstrates that all solutions in Y are not dominated by any solution in X . $SC(X, Y)$ is greater than $SC(Y, X)$ shows that the overall quality of the solutions in X is better than that in Y .

4.3. Program running environment

All algorithms in this article are coded in MATLAB R2018a and run on a computer with an Intel Core i5 2.21 GHz processor and 16.0 GB RAM.

4.4. Parameter settings for IMDJA

The parameters setting of IMDJA is based on our experimental analysis and related literature review. Based on our experimental observations, the following two key parameters are considered for tuning the IMDJA:

- (1) N : The population size.
- (2) $RoJE$: The ratio of the number of Jaya elements ($NoJE$) to the total number of elements (TE) in OSV. Using the $RoJE$ value, the $NoJE$ is estimated as follows:

$$NoJE = \lceil RoJE * TE \rceil \quad (38)$$

To achieve the best combination of the above parameters, the Taguchi DOE method (Montgomery, 2008) is applied for the instances JSPLFW-04 and JSPLFW-29. Combinations of different values of the key parameters are given in Table 2. The other parameters are set as follows: the JRPJ probability i.e., crossover probability (P_c) = 1.0, the mutation probability (P_m) = 0.1, and the number of generations = 200. IMDJA runs 20 times independently for each combination of parameters. The mean HV value is used to evaluate each parameter combination. The trend of each factor level obtained for JSPLFW-04 and JSPLFW-29 is exhibited in Fig. 6(a) and (b). Fig. 6 shows that both parameters significantly influence on the solutions of the IMDJA. It can be seen that the best results are obtained for the combination $N = 150$ and $RoJE =$

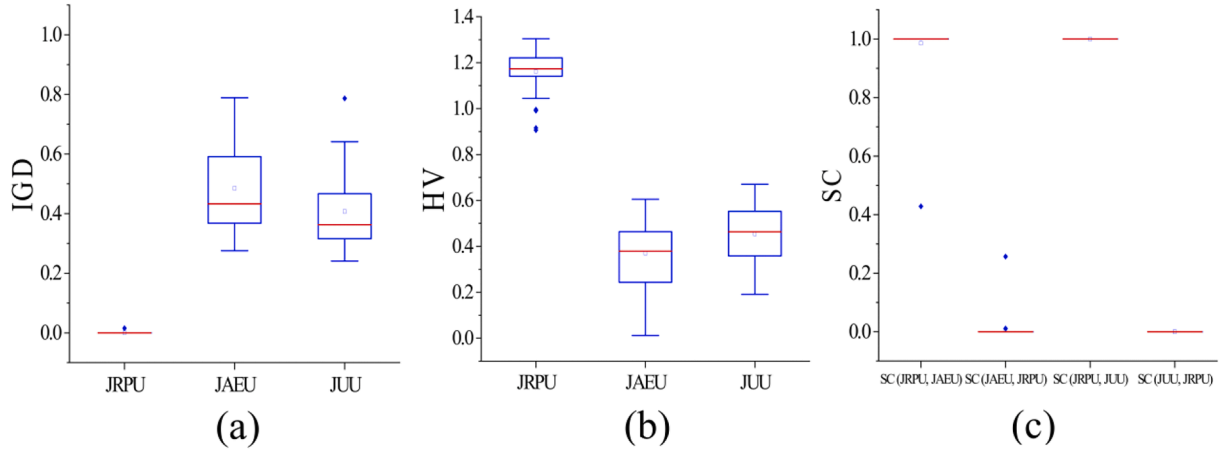


Fig. 7. Box plots for (a) IGD and (b) HV from Table 3 and (c) SC from Table 4.

Table 5

Mean value of IGD obtained by the IMDJA, NSGA-II, MOEA/D and mTLBO.

Instance	IGD			
	IMDJA	NSGA-II	MOEA/D	mTLBO
JSPLFW-01	0.0000	0.6364	0.5037	0.3439
JSPLFW-02	0.0000	0.5724	0.5166	0.2160
JSPLFW-03	0.0000	0.7335	0.6184	0.2858
JSPLFW-04	0.0034	0.4639	0.4542	0.2138
JSPLFW-05	0.0859	0.7399	0.7463	0.3755
JSPLFW-06	0.0081	0.5343	0.5846	0.2009
JSPLFW-07	0.0047	0.6128	0.5066	0.3246
JSPLFW-08	0.0353	0.6362	0.3094	0.2597
JSPLFW-09	0.0025	0.6615	0.6463	0.4631
JSPLFW-10	0.0000	0.9147	0.7326	0.5926
JSPLFW-11	0.0305	0.7425	0.6549	0.4164
JSPLFW-12	0.0143	0.4529	0.4426	0.3107
JSPLFW-13	0.0342	0.4339	0.6451	0.2569
JSPLFW-14	0.0147	0.4434	0.5271	0.2177
JSPLFW-15	0.0395	0.5060	0.4695	0.0273
JSPLFW-16	0.0387	0.5948	0.4624	0.3079
JSPLFW-17	0.0028	0.7826	0.8890	0.6186
JSPLFW-18	0.0386	0.6436	0.5396	0.3613
JSPLFW-19	0.0159	0.4413	0.4153	0.2731
JSPLFW-20	0.0201	0.4667	0.3951	0.3026
JSPLFW-21	0.0260	0.4388	0.3881	0.2071
JSPLFW-22	0.0267	0.4327	0.2068	0.1983
JSPLFW-23	0.0068	0.6445	0.7878	0.3693
JSPLFW-24	0.0194	0.4569	0.8595	0.3320
JSPLFW-25	0.0368	0.4972	0.5192	0.2281
JSPLFW-26	0.0591	0.6399	0.6307	0.4283
JSPLFW-27	0.0429	0.5494	0.4229	0.2318
JSPLFW-28	0.0386	0.6026	0.4840	0.1927
JSPLFW-29	0.0506	0.6538	0.3922	0.2460
JSPLFW-30	0.0118	0.7206	0.5682	0.4570
JSPLFW-31	0.0429	0.5414	0.6525	0.3690
JSPLFW-32	0.0967	0.5533	0.3108	0.2856
JSPLFW-33	0.0098	0.4528	0.3712	0.2527
JSPLFW-34	0.0447	0.5047	0.5010	0.2526
JSPLFW-35	0.0209	0.3988	0.5830	0.4582
JSPLFW-36	0.0243	0.5613	0.5041	0.4129
JSPLFW-37	0.0396	0.5908	0.5403	0.6210
JSPLFW-38	0.0299	0.4483	0.2855	0.2131
JSPLFW-39	0.0411	0.5042	0.2084	0.1920
JSPLFW-40	0.0142	0.3045	0.4054	0.3507

Table 6

Mean value of HV obtained by the IMDJA, NSGA-II, MOEA/D and mTLBO.

Instance	HV			
	IMDJA	NSGA-II	MOEA/D	mTLBO
JSPLFW-01	1.3108	0.3351	0.4275	0.6744
JSPLFW-02	1.2519	0.3514	0.3077	0.8260
JSPLFW-03	1.2585	0.1606	0.2668	0.6592
JSPLFW-04	1.2149	0.4123	0.3471	0.8054
JSPLFW-05	1.1135	0.1484	0.2129	0.6036
JSPLFW-06	1.1935	0.3434	0.2945	0.7912
JSPLFW-07	1.1801	0.2153	0.3203	0.6274
JSPLFW-08	1.1911	0.2201	0.5640	0.7149
JSPLFW-09	0.8675	0.1021	0.1864	0.3108
JSPLFW-10	1.2921	0.1721	0.3125	0.3826
JSPLFW-11	1.1893	0.1731	0.2346	0.4982
JSPLFW-12	1.0389	0.2873	0.3883	0.6786
JSPLFW-13	1.0554	0.3032	0.1632	0.6614
JSPLFW-14	1.0272	0.3105	0.3134	0.7520
JSPLFW-15	1.1301	0.3147	0.4786	1.1405
JSPLFW-16	0.9308	0.1243	0.2867	0.4859
JSPLFW-17	1.2115	0.1429	0.1012	0.3547
JSPLFW-18	1.1718	0.2154	0.2751	0.5644
JSPLFW-19	0.9612	0.2552	0.3935	0.6109
JSPLFW-20	1.0041	0.2712	0.6002	0.6269
JSPLFW-21	0.9667	0.3539	0.1020	0.7845
JSPLFW-22	1.1156	0.3874	0.7195	0.9301
JSPLFW-23	0.9749	0.2393	0.2093	0.6775
JSPLFW-24	0.9857	0.2637	0.1536	0.7206
JSPLFW-25	0.9523	0.2299	0.3320	0.6247
JSPLFW-26	0.9756	0.1393	0.2586	0.3971
JSPLFW-27	0.8890	0.1509	0.4030	0.5940
JSPLFW-28	1.0023	0.3617	0.5971	0.8303
JSPLFW-29	0.9374	0.1613	0.4589	0.6902
JSPLFW-30	0.7998	0.1171	0.2800	0.2969
JSPLFW-31	0.9128	0.2145	0.3877	0.6116
JSPLFW-32	0.8052	0.1702	0.4744	0.5398
JSPLFW-33	0.7965	0.2311	0.4903	0.6767
JSPLFW-34	0.8714	0.2138	0.3601	0.5682
JSPLFW-35	0.7087	0.2137	0.3041	0.6843
JSPLFW-36	0.8170	0.0650	0.3875	0.2565
JSPLFW-37	0.8291	0.1354	0.2400	0.3721
JSPLFW-38	0.9002	0.2514	0.6678	0.7066
JSPLFW-39	0.8955	0.3437	0.7082	0.6759
JSPLFW-40	0.6795	0.2030	0.5993	0.6448

0.05 for both the selected instances. Using the value of *RoJE*, the value of *NoJE* is evaluated and mentioned for each instance in Appendix 'A'.

4.5. Performance analysis of IMDJA for JSPLFW-ER

4.5.1. Effect of the number of Jaya elements

For IMDJA, Jaya random partial updating (JRPU) is proposed as discussed in Section 3.3. To show the significance of the JRPU mechanism, it is compared with Jaya all element updating (JAEU) mechanism

Table 7

Mean value of SC obtained by the IMDJA, NSGA-II, MOEA/D and mTLBO (IMDJA = J; NSGAII = N; MOEA/D = D; mTLBO = T).

Instance	SC					
	SC (J, N)	SC (N, J)	SC (J, D)	SC (D, J)	SC (J, T)	SC (T, J)
JSPLFW-01	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
JSPLFW-02	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
JSPLFW-03	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
JSPLFW-04	1.0000	0.0000	1.0000	0.0000	0.7727	0.0333
JSPLFW-05	1.0000	0.0000	1.0000	0.0000	0.3750	0.0000
JSPLFW-06	1.0000	0.0000	1.0000	0.0000	0.7568	0.0122
JSPLFW-07	1.0000	0.0000	1.0000	0.0000	0.5000	0.1176
JSPLFW-08	1.0000	0.0000	0.7500	0.0571	0.4167	0.0571
JSPLFW-09	1.0000	0.0000	0.8889	0.0000	0.5454	0.0511
JSPLFW-10	1.0000	0.0000	1.0000	0.0000	1.0000	0.0000
JSPLFW-11	1.0000	0.0000	1.0000	0.0000	0.4444	0.0000
JSPLFW-12	1.0000	0.0000	0.5238	0.0000	0.2307	0.0000
JSPLFW-13	1.0000	0.0000	0.3750	0.0000	0.2632	0.0000
JSPLFW-14	1.0000	0.0000	0.5179	0.0000	0.5238	0.0440
JSPLFW-15	1.0000	0.0000	0.6923	0.0000	0.3108	0.5732
JSPLFW-16	1.0000	0.0000	0.1154	0.0000	0.4400	0.0064
JSPLFW-17	1.0000	0.0000	1.0000	0.0000	0.9091	0.0000
JSPLFW-18	1.0000	0.0000	1.0000	0.0000	0.4642	0.0000
JSPLFW-19	1.0000	0.0000	0.3684	0.0000	0.4286	0.0000
JSPLFW-20	1.0000	0.0000	0.4286	0.0000	0.3809	0.0000
JSPLFW-21	1.0000	0.0000	0.1250	0.0000	0.2917	0.1011
JSPLFW-22	1.0000	0.0000	0.2857	0.0000	0.1481	0.0337
JSPLFW-23	1.0000	0.0000	0.5714	0.0000	0.2368	0.1078
JSPLFW-24	1.0000	0.0000	1.0000	0.0000	0.3571	0.0000
JSPLFW-25	1.0000	0.0000	0.1667	0.0000	0.2962	0.0093
JSPLFW-26	1.0000	0.0000	0.2000	0.0000	0.1739	0.0000
JSPLFW-27	1.0000	0.0000	0.5714	0.0000	0.2941	0.0000
JSPLFW-28	1.0000	0.0000	0.1250	0.0000	0.1429	0.0769
JSPLFW-29	1.0000	0.0000	0.1111	0.0208	0.1538	0.0208
JSPLFW-30	1.0000	0.0000	0.2000	0.0091	0.2857	0.0182
JSPLFW-31	1.0000	0.0000	0.1818	0.0000	0.1153	0.0000
JSPLFW-32	1.0000	0.0000	0.1429	0.0857	0.0284	0.2294
JSPLFW-33	1.0000	0.0000	0.4521	0.0000	0.1432	0.0000
JSPLFW-34	1.0000	0.0000	0.1111	0.0000	0.1600	0.0000
JSPLFW-35	1.0000	0.0000	0.2222	0.0051	0.0000	0.2704

Table 7 (continued)

Instance	SC					
	SC (J, N)	SC (N, J)	SC (J, D)	SC (D, J)	SC (J, T)	SC (T, J)
JSPLFW-36	1.0000	0.0000	0.1613	0.0125	0.2078	0.0125
JSPLFW-37	1.0000	0.0000	0.3415	0.0000	0.1923	0.0130
JSPLFW-38	1.0000	0.0000	0.3129	0.0000	0.3030	0.0308
JSPLFW-39	1.0000	0.0000	0.2098	0.0241	0.1333	0.0361
JSPLFW-40	1.0000	0.0000	0.2500	0.0000	0.1010	0.0000

for JA as adopted in [Caldeira and Gnanavelbabu \(2021\)](#) and also with Jaya uniform updating (JUU) (i.e. half of the elements of the individual are randomly selected to apply Jaya updating). Other parameter settings are the same for all three conditions. Accordingly, the performance of IMDJA with JRP, JAEU and JUU is compared based on the selected performance metrics. The algorithm is run for 20 times for all the conditions with 200 iterations in each run for all the instances. The results of IGD and HV are provided in [Table 3](#) and the results of SC are provided in [Table 4](#). The box plots of the results in [Table 3 and 4](#) are shown in [Fig. 7](#). It can be seen from the results that IMDJA with JRP performs much better than the JAEU and JUU.

4.5.2. Comparison with other algorithms

To further evaluate the effectiveness of the proposed algorithm, the performance of IMDJA is compared with well-known multi-objective optimization algorithms including NSGA-II ([Deb et al., 2002](#)), and MOEA/D ([Zhang and Li, 2007](#)). IMDJA is also compared with a multi-objective modified teaching learning-based optimization algorithm (mTLBO) ([Shen et al., 2016](#)) because the TLBO algorithm is similar with the Jaya algorithm. For a fair comparison, each algorithm's initial parameters are set as $N = 150$, $P_c = 1.0$, and $P_m = 0.1$, each algorithm runs 20 times for each instance, and the number of iterations is 200 in each run. Moreover, the mutation operation and external archive method are the same for all the algorithms as adopted for IMDJA. The other parameter settings for the compared algorithms are adopted the same as in the referred literature.

The results of IGD, HV, and SC are provided in [Table 5, 6 and 7](#) respectively. Regarding the IGD results in [Table 5](#), IMDJA performs better than NSGA-II and MOEA/D for all 40 instances. Compared to mTLBO, IMDJA performs better for 39 instances out of 40 instances.

Considering the HV results, IMDJA produces better results than both NSGA-II and MOEA/D for all 40 instances. While comparing with mTLBO, IMDJA produces better outcomes for 38 instances out of 40 instances.

Concerning the SC results, IMDJA outperforms for all 40 instances while comparing with NSGA-II and MOEA/D. For mTLBO, IMDJA results are better than mTLBO for 37 instances out of 40 instances.

The box plots of the results are shown in [Fig. 8](#). The results show that IMDJA is very competitive when compared with NSGA-II, MOEA/D and mTLBO algorithm for solving the JSPLFW-ER.

In addition to the above performance metrics, the superiority of the proposed IMDJA is also evaluated based on the total number of non-dominated solutions generated by each algorithm. The ratio of the number of obtained non-dominated solutions (nNS) between IMDJA and other compared algorithms is provided in [Table 8](#). The results indicate that the number of non-dominated solutions generated by IMDJA is obviously higher than that generated by the compared algorithms for most of the instances. The higher number of non-dominated solutions generated by IMDJA indicates a more diverse and comprehensive exploration of the search space than the compared algorithms, which provides greater flexibility in selecting the solution according to the

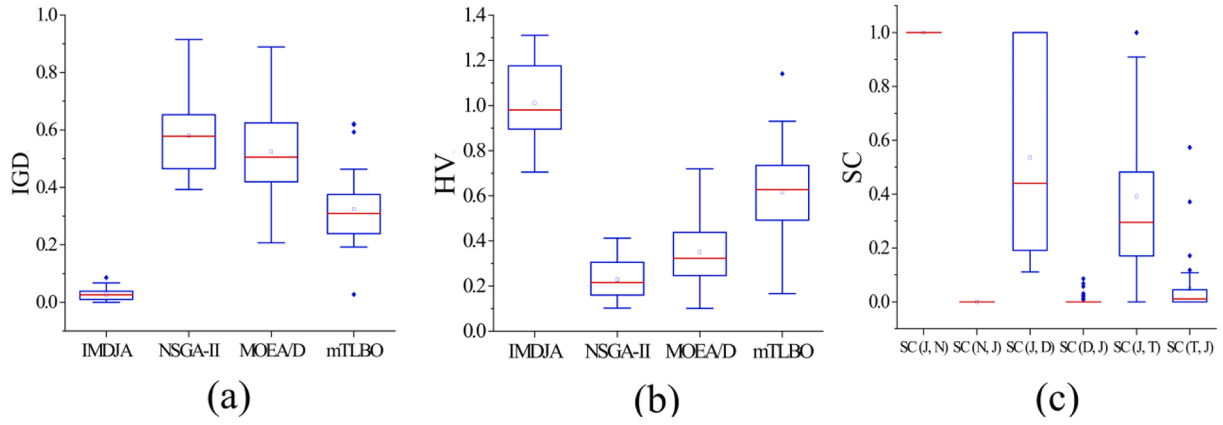


Fig. 8. Box plots for (a) IGD, (b) HV and (c) SC from Table 5, 6 and 7 respectively.

Table 8

The ratio of the number of non-dominated solutions (nNS) between IMDJA and compared algorithms (IMDJA = J; NSGAII = N; MOEA/D = D; mTLBO = T).

Instance	nNS_J / nNS_N	nNS_J / nNS_D	nNS_J / nNS_T
JSPLFW-01	1.29	3.60	4.5
JSPLFW-02	2.32	5.50	2.44
JSPLFW-03	3.85	8.33	5.00
JSPLFW-04	4.63	7.50	4.62
JSPLFW-05	1.21	5.17	4.29
JSPLFW-06	11.71	10.25	3.57
JSPLFW-07	4.55	7.14	2.17
JSPLFW-08	1.48	4.86	4.25
JSPLFW-09	7.61	12.45	7.21
JSPLFW-10	2.71	4.22	4.75
JSPLFW-11	2.00	5.25	2.33
JSPLFW-12	6.2	10.27	3.14
JSPLFW-13	2.70	6.08	1.74
JSPLFW-14	4.67	6.43	2.25
JSPLFW-15	3.15	9.11	4.10
JSPLFW-16	9.24	13.08	5.06
JSPLFW-17	12.67	15.20	6.33
JSPLFW-18	1.67	2.50	1.67
JSPLFW-19	3.07	6.36	2.78
JSPLFW-20	3.67	7.62	2.83
JSPLFW-21	1.85	4.36	1.41
JSPLFW-22	6.85	8.09	1.78
JSPLFW-23	9.82	23.86	10.44
JSPLFW-24	5.15	12.88	2.06
JSPLFW-25	4.65	11.89	3.34
JSPLFW-26	3.57	5.56	1.29
JSPLFW-27	5.45	5.45	2.73
JSPLFW-28	5.20	3.06	5.78
JSPLFW-29	3.43	6.86	4.00
JSPLFW-30	8.1	12.22	9.17
JSPLFW-31	3.78	8.50	2.83
JSPLFW-32	4.38	3.89	1.13
JSPLFW-33	10.75	3.58	1.48
JSPLFW-34	4.06	7.22	2.60
JSPLFW-35	9.33	15.08	10.32
JSPLFW-36	8.00	7.27	2.05
JSPLFW-37	4.53	8.56	2.66
JSPLFW-38	4.64	8.13	1.97
JSPLFW-39	2.96	3.19	2.44
JSPLFW-40	11.56	12.24	12.24

preference of the problem.

The mean computation time of IMDJA, NSGA-II, MOEA/D and mTLBO is presented in Table 9 to evaluate the efficiency of each algorithm. It can be seen from the table that the computing time of IMDJA is smaller than NSGA-II in 32 instances out of 40 instances. The computation time of IMDJA is higher than NSGA-II for the other 8 instances, which are from the very large size instances. When compared with MOEA/D and mTLBO, the computation time of IMDJA is much lower for

Table 9

Mean computation time of IMDJA, NSGA-II, MOEA/D and mTLBO.

Instance	Mean computation time (s)			
	IMDJA	NSGAII	MOEA/D	mTLBO
JSPLFW-01	186.22	198.53	314.78	1129.25
JSPLFW-02	174.66	181.25	315.92	1012.74
JSPLFW-03	129.34	147.16	229.37	794.45
JSPLFW-04	118.76	139.00	214.59	667.83
JSPLFW-05	125.47	140.00	237.20	797.71
JSPLFW-06	139.13	152.96	268.01	844.56
JSPLFW-07	125.89	141.89	239.30	756.00
JSPLFW-08	183.26	197.48	343.10	1101.38
JSPLFW-09	180.77	195.92	344.33	1098.76
JSPLFW-10	190.20	2011.37	325.19	1079.76
JSPLFW-11	158.38	158.46	250.84	828.45
JSPLFW-12	135.29	189.92	255.54	802.86
JSPLFW-13	198.94	211.18	370.21	1234.0
JSPLFW-14	139.86	152.33	265.77	836.21
JSPLFW-15	145.60	158.99	273.10	887.36
JSPLFW-16	143.50	157.13	274.50	879.92
JSPLFW-17	123.10	143.04	236.70	757.11
JSPLFW-18	90.28	111.34	160.10	519.23
JSPLFW-19	154.45	167.28	280.50	897.96
JSPLFW-20	156.54	166.83	282.20	908.79
JSPLFW-21	158.18	170.87	286.40	926.54
JSPLFW-22	162.80	174.09	294.10	957.00
JSPLFW-23	164.97	173.23	300.01	986.95
JSPLFW-24	145.19	157.90	272.40	834.37
JSPLFW-25	145.34	161.96	274.68	860.00
JSPLFW-26	195.10	207.46	328.00	1073.00
JSPLFW-27	187.17	199.01	306.30	1047.00
JSPLFW-28	167.34	178.51	274.04	927.34
JSPLFW-29	190.45	199.89	320.9	1059.58
JSPLFW-30	191.66	203.69	324.20	1062.00
JSPLFW-31	279.48	280.01	446.4	1492.01
JSPLFW-32	291.13	289.85	462.67	1575.30
JSPLFW-33	297.51	295.44	475.13	1573.15
JSPLFW-34	309.98	308.12	505.78	1719.99
JSPLFW-35	312.27	313.01	510.14	1711.00
JSPLFW-36	532.21	475.13	694.81	2454.96
JSPLFW-37	593.17	521.29	823.75	2885.00
JSPLFW-38	599.79	543.1	855.88	2937.26
JSPLFW-39	637.7	571.32	924.65	3206.42
JSPLFW-40	615.41	552.30	881.60	3132.63

all 40 instances.

To assess the presence of differences between the outcomes of the proposed IMDJA and the outcomes of other compared algorithms, the t -test is employed, with a significance level set at 0.05. The statistical tests are performed separately for IMDJA and NSGA-II, IMDJA and MOEA/D, and IMDJA and mTLBO. The p -value ≤ 0.05 , indicates a significant difference between the outcomes of IMDJA and the compared algorithm. Conversely, the p -value > 0.05 , indicates no significant difference

Table 10

T-test results between IMDJA, and NSGA-II, MOEA/D and mTLBO.

Instance	p-value		
	IMDJA - NSGAII	IMDJA - MOEA/D	IMDJA - mTLBO
JSPLFW-01	7.69e-11	3.53e-07	1.54e-11
JSPLFW-02	5.00e-14	0.046	0.4015
JSPLFW-03	7.78e-12	0.6042	0.0587
JSPLFW-04	9.13e-10	0.3363	0.0109
JSPLFW-05	1.06e-9	0.0130	4.11e-4
JSPLFW-06	8.83e-22	0.1072	7.32e-5
JSPLFW-07	9.47e-12	0.0120	1.46e-06
JSPLFW-08	9.25e-8	1.61e-4	9.04e-03
JSPLFW-09	0.2318	0.0015	1.66e-4
JSPLFW-10	1.15e-14	4.94e-12	3.68e-11
JSPLFW-11	3.03e-11	0.012	2.51e-05
JSPLFW-12	31.22e-18	0.005	2.81e-08
JSPLFW-13	8.50e-10	0.0023	2.01e-06
JSPLFW-14	2.09e-11	2.68e-06	4.75e-07
JSPLFW-15	1.61e-10	1.06e-05	0.0012
JSPLFW-16	0.1652	6.68e-48	2.54e-16
JSPLFW-17	4.18e-11	0.0471	2.10e-04
JSPLFW-18	4.59e-12	0.0013	0.0019
JSPLFW-19	1.59e-09	2.85e-4	2.59e-09
JSPLFW-20	8.21e-15	4.65e-13	3.23e-10
JSPLFW-21	9.89e-11	2.71e-5	9.82e-06
JSPLFW-22	3.09e-22	5.78e-07	2.43e-14
JSPLFW-23	0.0304	3.40e-07	7.84e-10
JSPLFW-24	1.65e-11	0.0108	1.68e-9
JSPLFW-25	1.41e-15	7.15e-7	2.81e-16
JSPLFW-26	3.44e-7	3.57e-5	9.01e-17
JSPLFW-27	1.48e-6	1.15e-11	6.67e-12
JSPLFW-28	2.21e-13	2.53e-8	9.55e-07
JSPLFW-29	6.88e-11	1.39e-07	5.29e-09
JSPLFW-30	0.0491	2.30e-09	3.67e-6
JSPLFW-31	1.40e-10	1.231e-13	1.30e-32
JSPLFW-32	1.28e-8	6.92e-9	6.53e-39
JSPLFW-33	0.0016	7.78e-07	1.58e-20
JSPLFW-34	4.60e-14	1.36e-16	1.67e-19
JSPLFW-35	1.77e-12	4.91e-14	2.29e-18
JSPLFW-36	1.58e-11	1.64e-17	6.56e-24
JSPLFW-37	1.79e-20	1.75e-17	2.95e-27
JSPLFW-38	3.42e-14	0.0014	1.62e-9
JSPLFW-39	9.88e-24	4.25e-17	1.19e-20
JSPLFW-40	1.04e-09	7.11e-12	8.75e-34

between the IMDJA and the compared algorithm. The t -test results are provided in Table 10, with bolded results indicating a significant difference between the outcomes of the two algorithms. Table 10 shows that there is a significant difference between IMDJA and both NSGA-II and mTLBO in 38 instances out of 40 instances. Similarly, there is a significant difference between IMDJA and MOEA/D in 37 instances out of 40 instances.

4.6. Case study and analysis

4.6.1. Description of an industrial case

An industrial case of marble goods production from Indus Marble (<https://www.indusmarble.com>) is considered to investigate the proposed JSPLFW-ER approach. The facility can produce a variety of customized products in a job-shop environment. There are five jobs $J = \{J_1, J_2, J_3, J_4, \text{ and } J_5\}$, five machines $M = \{M_1, M_2, M_3, M_4, \text{ and } M_5\}$ and three workers $W = \{W_1, W_2, \text{ and } W_3\}$, making the instance $5 \times 5 \times 3$. Workers are totally flexible, i.e. each worker can perform every machine. However, the machines are customized and each machine is dedicated to perform a specific operation. Accordingly, the operations of cutting, grinding, engraving, polishing, and fitment are done by M_1, M_2, M_3, M_4 , and M_5 , respectively. Each job has a certain number of operations to be performed on the designated machines in the prescribed processing sequence. The processing information of each job, processing time, and the human energy expenditure of each worker for the respective operation are given in Table 11. For example, the processing

Table 11Processing time and human energy expenditure of each worker for the case study ($5 \times 5 \times 3$).

Job	Operation	Machine	Worker	Process time (min)	Human energy expenditure (Kcal)
J_1	O_{11}	M_1	W_1, W_2, W_3	26, 32, 38	150, 162, 173
	O_{12}	M_2	W_1, W_2, W_3	36, 30, 43	180, 169, 191
	O_{13}	M_3	W_1, W_2, W_3	28, 25, 20	97, 86, 70
	O_{14}	M_4	W_1, W_2, W_3	41, 33, 43	202, 189, 207
	O_{15}	M_5	W_1, W_2, W_3	17, 16, 14	54, 51, 45
J_2	O_{21}	M_1	W_1, W_2, W_3	26, 32, 38	150, 162, 173
	O_{22}	M_2	W_1, W_2, W_3	36, 30, 43	180, 169, 191
	O_{23}	M_4	W_1, W_2, W_3	41, 33, 43	202, 189, 207
J_3	O_{31}	M_1	W_1, W_2, W_3	26, 32, 38	150, 162, 173
	O_{32}	M_2	W_1, W_2, W_3	36, 30, 43	180, 169, 191
	O_{33}	M_3	W_1, W_2, W_3	28, 25, 20	97, 86, 70
	O_{34}	M_4	W_1, W_2, W_3	41, 33, 43	202, 189, 207
J_4	O_{41}	M_2	W_1, W_2, W_3	30, 24, 32	137, 125, 140
	O_{42}	M_3	W_1, W_2, W_3	25, 23, 21	94, 89, 77
	O_{43}	M_4	W_1, W_2, W_3	36, 28, 39	163, 150, 169
	O_{44}	M_5	W_1, W_2, W_3	17, 16, 16	54, 52, 52
J_5	O_{51}	M_2	W_1, W_2, W_3	30, 24, 32	137, 125, 140
	O_{52}	M_4	W_1, W_2, W_3	36, 28, 39	163, 150, 169

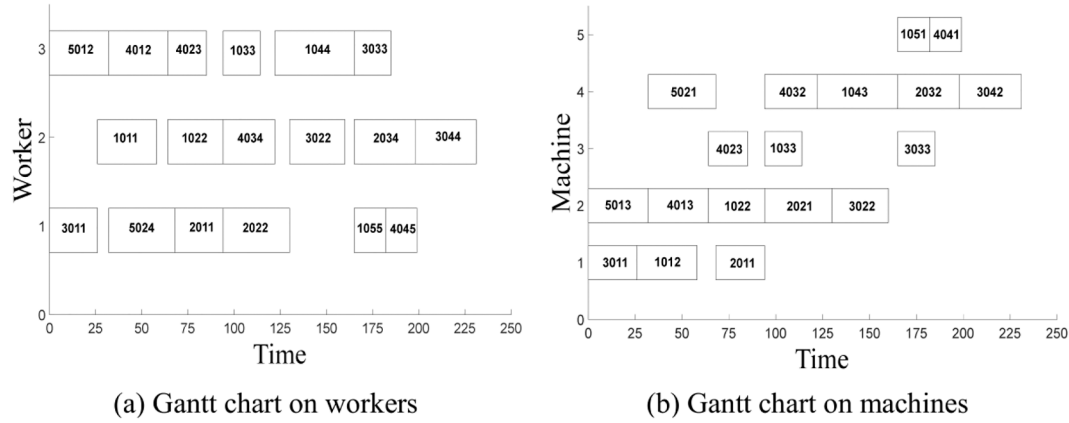
time for O_{11} by W_1, W_2 and W_3 is 26, 32, and 38 respectively and the human energy expenditure is 150, 162 and 173 respectively. The human energy expenditure to perform an operation is calculated based on the energy equations defined in Section 2.3.1. Moreover, the critical limit of the average human energy expenditure rate is set as $\bar{E}_{lim} = 5.0$ Kcal/min. The formulation for calculating the total human energy expenditure of an operation is developed based on Garg's method (Garg et al. 1978), and provided in Appendix B. As an example, the calculation for the total energy expenditure of W_1, W_2 and W_3 for all the operations of job 1 of the case study is provided in Appendix C. The Appendix-B and Appendix-C are provided on https://www.researchgate.net/publication/374417627_Appendices-ABC with the extraction code: asir49.

4.6.2. Experiment 1

The proposed JSPLFW-ER approach is tested with IMDJA based on the above parameter settings for the industrial problem described in Table 11. The IMDJA runs 40 times to obtain the non-dominated solutions before applying the constraint at Eq. (32). Only twenty randomly selected non-dominated solutions are provided in Table 12 for evaluation. The non-dominated solutions at solution number (SN) 8, 14, and 20 are not feasible as the value of $\bar{E}_s$ is beyond the $\bar{E}_{lim}$. Constraint at Eq. (32) is applied to remove these solutions. The value of $\bar{E}_s$ for other non-dominated solutions is under the critical limit, therefore, considering the appropriate value of optimization objectives, i.e. makespan (CT_M), maximum worker workload (TO_M), and maximum human energy expenditure (EO_M), any non-dominated solution among them can be selected. For example, prioritizing the makespan, SN 16 with the minimum value of makespan among the feasible solutions can be considered the most suitable solution, as shown in Table 12.

Table 12Part of the results by IMDJA for JSPLFW-ER for experiment 1 (Instance $5 \times 5 \times 3$).

SN	CT_M	TO_M	EO_M	TI_s	EI_s	$\bar{E}_s$	SN	CT_M	TO_M	EO_M	TI_s	EI_s	$\bar{E}_s$
1	246	196	867	50	90	3.88	11	249	197	851	52	94	3.78
2	256	193	847	63	113	3.73	12	228	171	911	57	103	4.43
3	278	179	854	99	178	3.69	13	254	197	841	57	103	3.70
4	244	172	908	72	130	4.23	14	210	205	1136	5	9	5.45
5	347	173	886	174	313	3.42	15	227	209	886	18	32	4.04
6	246	196	867	50	90	3.88	16	220	172	947	48	86	4.68
7	256	193	847	63	113	3.73	17	229	197	866	32	58	4.02
8	212	203	1117	9	16	5.34	18	248	186	858	62	112	3.89
9	235	191	887	44	79	4.10	19	244	197	861	47	85	3.86
10	330	185	851	145	261	3.34	20	216	200	1097	16	29	5.21

**Fig. 9.** Gantt charts illustrating a solution for the instance $5 \times 5 \times 3$ of JSPLFW-ER.**Table 13**Part of the results by IMDJA for JSPLFW-ER for experiment 2 (Instance $5 \times 5 \times 2$).

SN	CT_M	TO_M	EO_M	TI_s	EI_s	$\bar{E}_s$	SN	CT_M	TO_M	EO_M	TI_s	EI_s	$\bar{E}_s$
(a) W_1 and W_2 available													
1	282	256	1250	26	47	4.59	11	251	251	1292	0	0	5.15
2	267	267	1247	0	0	4.67	12	321	255	1250	66	119	4.25
3	251	251	1292	0	0	5.15	13	267	267	1247	0	0	4.67
4	262	250	1335	12	22	5.17	14	252	252	1288	0	0	5.11
5	300	259	1246	41	74	4.39	15	260	260	1262	0	0	4.85
6	269	254	1272	15	27	4.82	16	280	255	1251	25	45	4.62
7	276	250	1269	26	47	4.76	17	277	249	1338	28	50	5.00
8	265	258	1251	7	13	4.77	18	271	255	1252	16	29	4.72
9	299	249	1337	50	90	4.76	19	263	256	1261	7	13	4.84
10	358	258	1247	100	180	3.97	20	282	252	1253	30	54	4.63
(b) W_1 and W_3 available													
1	295	284	1360	11	20	4.67	11	297	281	1338	16	29	4.60
2	346	295	1273	51	92	3.93	12	302	275	1366	27	49	4.68
3	309	279	1349	30	54	4.53	13	311	290	1284	21	38	4.25
4	322	289	1299	33	59	4.21	14	330	298	1273	32	58	4.02
5	318	285	1316	33	59	4.32	15	351	286	1301	65	117	4.03
6	316	275	1344	41	74	4.48	16	302	289	1300	13	23	4.38
7	343	279	1327	64	115	4.19	17	342	284	1316	58	104	4.14
8	327	287	1304	40	72	4.20	18	308	292	1291	16	29	4.28
9	293	293	1283	0	0	4.38	19	298	289	1326	9	16	4.50
10	307	274	1383	33	59	4.69	20	311	287	1326	24	43	4.40
(c) W_2 and W_3 available													
1	293	280	1300	13	23	4.51	11	284	276	1420	8	14	5.05
2	297	267	1334	30	54	4.67	12	355	266	1350	89	160	4.24
3	340	281	1283	59	106	4.07	13	287	287	1411	0	0	4.92
4	314	281	1296	33	59	4.31	14	322	296	1267	26	47	4.07
5	274	274	1442	0	0	5.26	15	304	264	1392	40	72	4.81
6	318	284	1286	34	61	4.23	16	301	276	1314	25	45	4.51
7	315	273	1330	42	76	4.45	17	305	272	1333	33	59	4.56
8	295	265	1404	30	54	4.93	18	285	274	1418	11	20	5.04
9	327	279	1295	48	86	4.21	19	327	294	1274	33	59	4.07
10	348	287	1261	61	110	3.93	20	303	286	1279	17	31	4.32

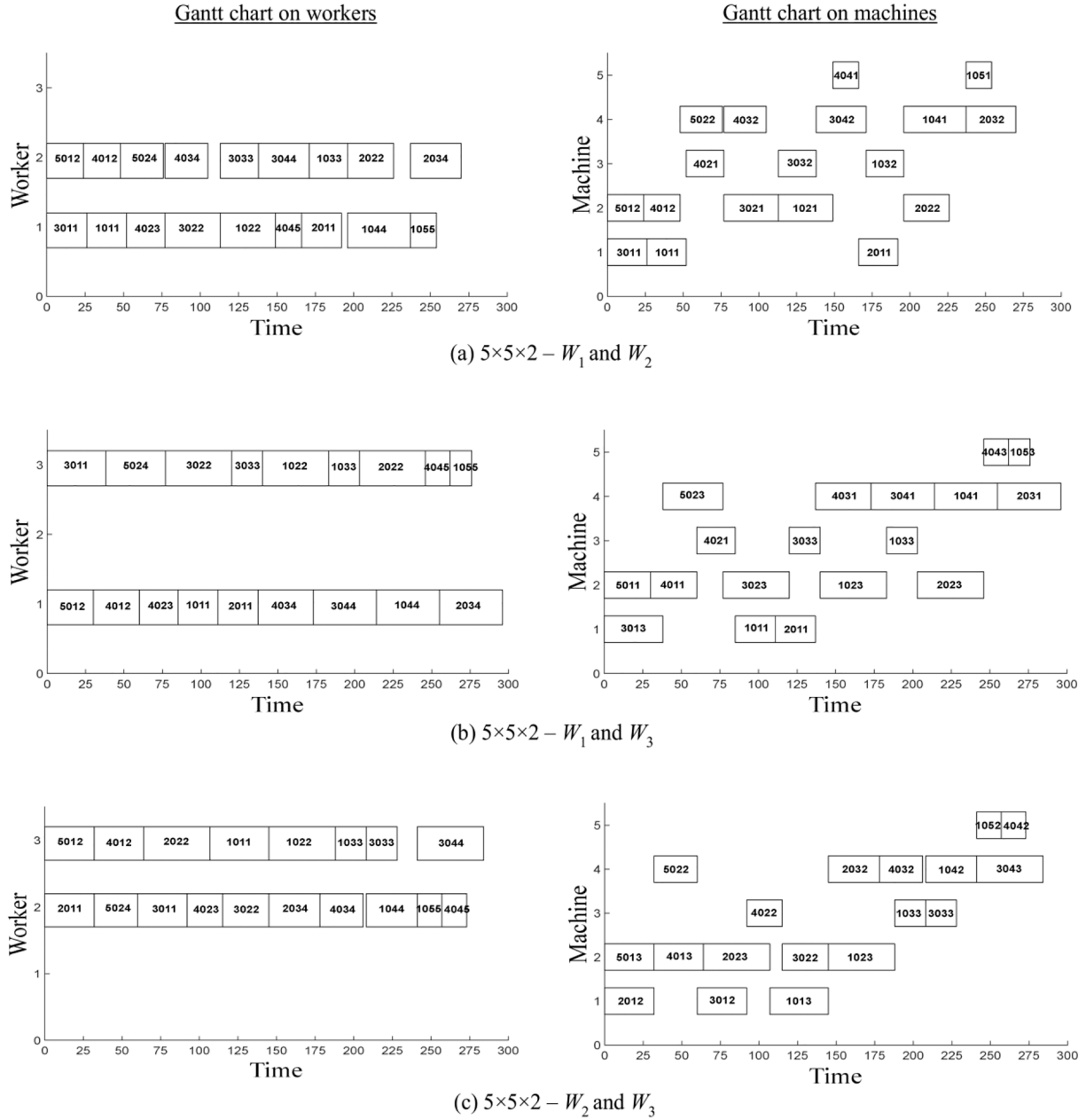


Fig. 10. Gantt charts illustrating the solutions for the sub-instances of $5 \times 5 \times 2$.

Table 14

Results of experiment 3 (Instance $5 \times 5 \times 3$ -JSP without considering worker flexibility).

SN	CT_M	TO_M	EO_M	TI_s	EI_s	$\bar{E}_s$
1	293	293	1624	0	0	5.54

The Gantt chart of a solution is illustrated in Fig. 9. As shown in Fig. 9 (a), the processing sequence represents the processing operation sequence of different jobs by different workers. For example, for worker W_1 the processing sequence is 3011-5024-2011-2022-1055-4045. It consists of different processing operations of the jobs, where “5024” means that the second processing operation of job J_5 is done on machine 4. Similarly, Fig. 9(b) shows the processing operation sequence of different jobs on different machines. For example, for machine M_3 , the processing sequence is 4023-1033-3033, where “4023” means that the

second processing operation of job J_4 is done by worker 3. Moreover, the calculated total workload TO_s of W_1 , W_2 , and W_3 is 171, 174, 174 (min) respectively; and total human energy expenditure E_s (i.e., $EO_s + EI_s$) of W_1 , W_2 , and W_3 is 957, 998, 838 (Kcal/min) respectively. It can be seen that the workload and human energy expenditure are well balanced amongst all the workers.

4.6.3. Experiment 2

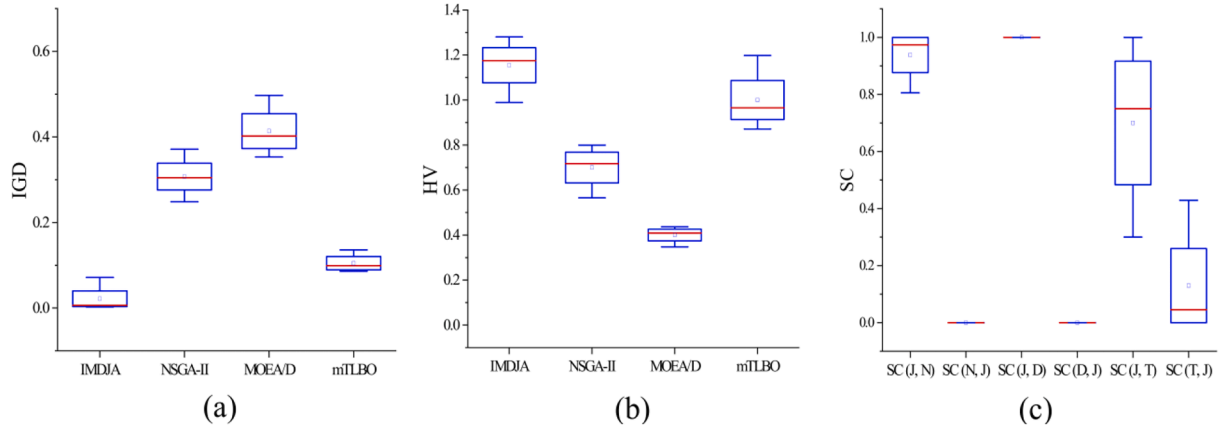
The proposed JSPLFW-ER approach is tested when only two workers are available to perform operations for the industrial problem provided in Table 11, making the instance $5 \times 5 \times 2$. The probable sub-instances of $5 \times 5 \times 2$ are:

- (a) $5 \times 5 \times 2 - W_1$ and W_2 .
- (b) $5 \times 5 \times 2 - W_1$ and W_3 .
- (c) $5 \times 5 \times 2 - W_2$ and W_3 .

Table 15

Mean value of IGD, HV and SC obtained by the IMDJA, NSGA-II, MOEA/D and mTLBO (IMDJA = J; NSGAII = N; MOEA/D = D; mTLBO = T).

Instance	IGD					
	IMDJA	NSGA-II	MOEA/D	mTLBO		
$5 \times 5 \times 3 (W_1, W_2, W_3)$	0.0086	0.3713	0.3927	0.1358		
$5 \times 5 \times 2 (W_1, W_2)$	0.0025	0.3057	0.4972	0.1051		
$5 \times 5 \times 2 (W_1, W_3)$	0.0042	0.2483	0.3533	0.0858		
$5 \times 5 \times 2 (W_2, W_3)$	0.0715	0.3039	0.4116	0.0927		
	HV					
	IMDJA	NSGA-II	MOEA/D	mTLBO		
$5 \times 5 \times 3 (W_1, W_2, W_3)$	1.1852	0.5654	0.4363	0.8712		
$5 \times 5 \times 2 (W_1, W_2)$	1.2802	0.7992	0.4167	1.1981		
$5 \times 5 \times 2 (W_1, W_3)$	1.1643	0.7369	0.3474	0.9557		
$5 \times 5 \times 2 (W_2, W_3)$	0.9891	0.6981	0.4008	0.9750		
	SC					
	SC (J, N)	SC (N, J)	SC (J, D)	SC (D, J)	SC (J, T)	SC (T, J)
$5 \times 5 \times 3 (W_1, W_2, W_3)$	0.9474	0.0000	1.0000	0.0000	1.0000	0.0000
$5 \times 5 \times 2 (W_1, W_2)$	1.000	0.0000	1.0000	0.0000	0.6667	0.0909
$5 \times 5 \times 2 (W_1, W_3)$	1.000	0.0000	1.0000	0.0000	0.8333	0.0000
$5 \times 5 \times 2 (W_2, W_3)$	0.8056	0.0000	1.0000	0.0000	0.3000	0.4286

**Fig. 11.** Box plots for (a) IGD, (b) HV and (c) SC from Table 15 of case study instances.

IMDJA runs 40 times without applying the constraint at Eq. (32) for all the sub-instances of $5 \times 5 \times 2$ to obtain the non-dominated solutions, from which, only twenty randomly selected non-dominated solutions are provided in Table 13 for each sub-instance. Table 13(a) shows the solutions when only workers W_1 and W_2 are available. In this case, the solutions at SN 3, 4, 11, 14, and 17 are not feasible as the value of $\bar{E}_s$ is beyond $\bar{E}_{lim}$. The other solutions are considered appropriate as the value of $\bar{E}_s$ is under the critical limit. Table 13(b) shows the solutions when workers W_1 and W_3 are available. In this case, all the solutions are feasible as the value of $\bar{E}_s$ for all the solutions is under the critical limit. Table 13(c) shows the optimal values of objective functions when only workers W_2 and W_3 are available. Results show that the solutions at SN 5, 11, and 18 are not feasible as the value of $\bar{E}_s$ is beyond $\bar{E}_{lim}$ for these solutions. The other solutions are considered feasible as the value of $\bar{E}_s$ is under the critical limit. The Gantt charts for each sub-instance of $5 \times 5 \times 2$ are illustrated in Fig. 10.

Results of experiments 1 and 2 reveal that consideration of ergonomic factors is important for the manufacturing industries having limited flexible workers to balance the load among the workers and avoid excessive load on a single worker. Minimizing the TO_M is helpful in balancing the workload among workers to deal with the limited worker problems. Similarly, minimizing the EO_M is helpful in balancing the human energy expenditure and is critical to prevent exorbitant human energy expenditure by any of the workers. Moreover, the value of $\bar{E}_s$ is

also useful to satisfy the critical limit of the average human energy expenditure rate while selecting the appropriate solution from the obtained non-dominated solutions. Solutions with a value of $\bar{E}_s$ beyond the critical limit are omitted.

4.6.4. Experiment 3

In this experiment, job-shop scheduling is carried out for instance $5 \times 5 \times 3$ provided in Table 11 without considering the worker flexibility. The worker who is most efficient to perform an operation is considered to execute the respective operation. The information can be obtained from Table 11. For this case, the obtained solution is provided in Table 14. The results show that the obtained value of makespan is much higher than that of many optimal solutions obtained in experiment-1. Furthermore, the value of $\bar{E}_s$ is also higher than $\bar{E}_{lim}$. Results of experiment 3 show that worker flexibility is not only important to minimize the makespan of a JSP but also very useful to relieve the workers.

The comparison of results from experiments 1 and 3 demonstrates the significance of worker flexibility. In experiment-1, the workload is effectively balanced among the workers through utilizing the worker flexibility. Additionally, experiment-1 produces a number of non-dominated solutions with significantly lower makespan values, along with corresponding values of $\bar{E}_s$ are also much lower than those obtained in experiment-3. The solutions obtained in experiment-1 are helpful for the factory manager in selecting the appropriate solution.

4.6.5. Other results and discussion

Each of the algorithms, IMDJA, NSGA-II, MOEA/D and mTLBO, runs 20 times with 200 iterations in each run for the instances in experiment 1 and 2. The results are provided in Table 15 and the boxplots of IGD, HV and SC achieved by each algorithm are shown in Fig. 11. It can be seen from the result that the proposed IMDJA performs better than the NSGA-II, MOEA/D and mTLBO when solving the real-world industrial case study.

5. Conclusion and future studies

In this paper, job-shop scheduling with limited flexible workers considering the ergonomic factors (JSPLFW-ER) is investigated with the objective to minimize the makespan, maximum worker workload, and maximum human energy expenditure of the assigned operations. The average human energy expenditure rate is also evaluated to ensure that it does not exceed the critical limit. An improved multi-objective discrete Jaya algorithm (IMDJA) is employed to solve JSPLFW-ER. A Jaya random partial updating is proposed for IMDJA, i.e., instead of considering all the elements of the individual for Jaya updating, a certain number of elements of the individual are randomly selected to apply Jaya updating. The number of selected elements of the individual for Jaya updating is evaluated using the Taguchi DOE method. The performance of IMDJA is investigated using the 40 newly constructed instances and found competitive to solve JSPLFW-ER while comparing with other algorithms.

The proposed JSPLFW-ER approach is applied to the real-world industrial case study. Multiple experiments are performed with different instances concerning the limited flexible workers. The results show that the JSPLFW-ER approach aids the factory manager in selecting an appropriate solution from the obtained optimal solutions with simultaneously considering makespan, maximum worker workload, and maximum human energy expenditure. Moreover, the proposed IMDJA is also compared with other algorithms and found competitive for solving the various instances of the case study. JSPLFW-ER approach can be applied to the job-shop scheduling problems having limited flexible workers to prioritize the productivity and alleviate the workforce, particularly when recurring manual work is involved.

In the proposed JSPLFW-ER, transportation time and set-up time are not considered. However, in some industrial problems, the consideration of these factors is important and cannot be neglected. Moreover, in our model only one worker is considered to perform an operation on a machine at a time. However, there are cases where more than one worker may be required to perform complex operations on some machines. In the future work, we will consider these aspects to be included in the proposed model. Moreover, the ergonomic consideration will be extended for flexible Job-shop scheduling problem with limited flexible workers.

CRediT authorship contribution statement

Shaban Usman: Conceptualization, Methodology, Software, Formal analysis, Validation, Data curation, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Cong Lu:** Conceptualization, Supervision, Validation, Investigation, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have shared the link to data in the manuscript.

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